Prostate Cancer

Condensed rapid revision · facts already answered in Papers 1–4 point to that answer

highlighted number to memorise · **KEY POINT** and **ALGORITHM** boxes carry the reasoning · MCQ exam trivia · † added beyond the source notes · P2 topic = covered in full in that Paper's answer book

Chapter	Topic
Ch 153	Epidemiology, Aetiology & Prevention
Ch 154	Prostate Cancer Biomarkers
Ch 155	Prostate Biopsy
Ch 156	Pathology of Prostatic Neoplasia
Ch 157	Diagnosis & Staging
Ch 158	Active Management of Localised Disease
Ch 159	Active Surveillance
Ch 160	Open Radical Prostatectomy
Ch 161	Laparoscopic & Robotic Prostatectomy, PLND
Ch 162	Radiation Therapy
Ch 163	Focal Therapy
Ch 164	Locally Advanced Prostate Cancer
Ch 165	Recurrent & Newly Metastatic Disease
Ch 166	Castration-Resistant Prostate Cancer
★	Putting it together

153 Epidemiology, Aetiology and Prevention of Prostate Cancer

1 · Epidemiology — the headline

- Worldwide: **second commonest** non-skin cancer, **fifth** cause of cancer death in men; 2020 ≈ **1.4 million cases, 375,000 deaths**.
- Incidence varies ~13-fold (South Central Asia lowest, Northern Europe highest); US lifetime risk ≈ **1 in 8**; 5-year survival 97%, yet second cause of cancer death in US men.
- **Incidence and mortality are decoupled.** Incidence tracks screening and wealth; mortality peaks where African ancestry meets poor access — **Caribbean has the highest mortality in the world** MCQ. Asia lowest on both, but rising.
- **Latent (autopsy) cancer**: ~5% <30 y → 29% at 60–69 → **59% >79 y**; higher in Black men at every age.

> **KEY POINT**
> Aetiology is hard to study: in a PSA-screened population a diagnosis mostly means a man "happened to have a PSA test", so risk-factor studies may find determinants of *being tested*. Preferred end point: **aggressive disease** (advanced, metastatic, death). **Healthy-user bias**: health-conscious men seek PSA tests, so healthy habits spuriously raise incidence while lowering mortality.

2 · Established risk factors

Only three are unequivocal: **age, race (ancestry) and family history**.

- **Age**: mean at diagnosis 66; ~60% at ≥65; rare <40. Diagnosis at ≥70 carries higher risk of cancer death.
- **Race**: Black men have highest incidence (172.6/100,000 US) and **twice** the cancer-specific mortality of White men, younger and more metastatic at presentation. Drivers: unequal access, comorbidity and **8q24** (and 17q21) alleles. **Equal access → equal outcome** in non-metastatic disease.
- **Family history**: ~20% of patients; twin heritability up to **58%** — genes outweigh shared lifestyle.

Family history	Relative risk
Any first-degree relative	2–3×
Two first-degree relatives	~3.5×
Both father and brother	**9×**
Relative diagnosed <60 vs ≥60	~2.5× vs ~1.6×
Father/brother died of prostate cancer	2× risk of prostate cancer death

3 · Germline genetics

Key inherited genes are **DNA-repair genes**; they predict **aggressive** cancer.

Gene	Pathway	Association	Implication
BRCA2, BRCA1	Homologous recombination (HR)	BRCA2 familial, aggressive	Early screening; PARP inhibitor; caution in AS

Gene	Pathway	Association	Implication
ATM	HR	Metastatic disease	As for BRCA
PALB2, CHEK2	HR / DNA repair	Familial; PALB2 lethal	CHEK2: targeted screening
MLH1, MSH2, MSH6, PMS2 (**Lynch**)	Mismatch repair (MMR)	2–5% of cases	Checkpoint inhibitor; very ADT-sensitive
HOXB13	Prostate-specific homeobox	Hereditary prostate cancer	Targeted screening
8q24, 17q21	GWAS loci	Commoner in African ancestry	Part of racial excess

- **Polygenic risk scores** predict *any* cancer, not aggressive cancer — not yet practice-changing.
- P4 Indications of genomic testing in carcinoma prostate — *germline testing: metastatic, high-risk, intraductal/cribriform, family history, Ashkenazi*

4 · Molecular aetiology and inflammation

- **Chronic inflammation**: near-universal in biopsies; raises PSA, but evidence supports a causal role.
- † **Proliferative inflammatory atrophy (PIA)**: proposed precursor, via HGPIN to cancer.
- **TMPRSS2-ERG** fusion: androgen-regulated promoter drives ERG (ETS family); ~half of Western tumours, uncommon in East Asians (FOXA1 mutations commoner in Chinese men). → *Ch 154*

5 · Possible risk factors — body, metabolism, lifestyle

Most lifestyle factors act on **progression** rather than initiation; evidence observational.

Factor	Total incidence	Aggressive / fatal
Height	~5% ↑ per 10 cm	↑ high-grade and death
Obesity	Mixed; ↓ low-grade	**Consistently ↑**
Smoking	Not ↑ (less screening)	**↑ ~40% mortality**; no excess after quitting ≥10 y
Metabolic syndrome, hypertension, CVD	Slight ↑	Metabolic syndrome: worse pathology
Diabetes	↓ (low testosterone, IGF-1, PSA; metformin)	—
Vigorous activity	No clear effect	↓ advanced disease
Alcohol	Unclear	↑ mortality ≥45 g/day; binge ↑

- **Mechanism**: adiposity → ↑ insulin/IGF-1, leptin, inflammation, oestrogen, ↓ SHBG → ↑ proliferation, ↓ apoptosis.
- **Adult serum testosterone does *not* predict risk** MCQ; genetically higher free testosterone, IGF-1 and early puberty modestly do.
- **Composite healthy lifestyle** → **30–50%** lower lethal cancer; in highest polygenic-risk quartile lifetime lethal risk 5.3% → 1.6%.
- ADT: baseline cardiovascular assessment (CVD link observational only).

- **Diet**: WCRF 2018 — **no dietary factor with strong evidence**. PSA-era downgrades: dairy/calcium → "limited"; lycopene → "no conclusion" (lycopene available only from cooked tomato in oil).
- **Vasectomy**: not a risk factor (detection bias); AUA: no need to discuss.
- Transgender women on oestrogen: median PSA ≈ **0.02 ng/mL** (50-fold lower) → lower the cut-points.

6 · Supplements

- P4 Antioxidants in urological malignancy; Prostate cancer prevention trials — *SELECT null, vitamin E harmful; VITAL null; lycopene, green tea, soy/pomegranate/curcumin*
- No supplement benefits healthy men on a balanced diet. Calcium supplements ↑ fatal cancer (caution on ADT); zinc ≥75 mg/day or ≥15 y → ↑ lethal.

> **KEY POINT**
> Vitamin E worked in ATBC but failed in SELECT: ATBC used a tenfold lower dose in **heavy smokers** with little PSA screening (aggressive cancers); SELECT used 400 IU in non-smokers with widespread screening (indolent cancers). Different disease, different answer.

7 · Chemoprevention

- P4 Prostate cancer prevention trials — *5ARI rationale, PCPT/REDUCE high-grade controversy, FDA warning, no mortality effect; aspirin, statins, metformin*
- No FDA-approved preventive agent; the bar is now prevention of **aggressive** cancer. 5ARIs shrink gland ~25%; on a 5ARI **double the PSA**. *→ Ch 154*
- **Secondary prevention** (Prostate Cancer Lifestyle Trial, men on AS): plant-based diet, exercise, stress management → **5% vs 27%** treated at 2 y.
- Activity ↓ cancer-specific mortality ~31% (≥150 min/week moderate or ≥75 min vigorous). Western diet ↑ cancer death (HR 2.5); high-fat dairy ↑ recurrence; continued smoking ↑ recurrence and death.

> **KEY POINT**
> Cardiovascular disease kills more men with non-metastatic prostate cancer than the cancer does — lifestyle advice benefits both.

8 · PSA screening

- P2 PSA screening for prostate cancer: evidence, problems and solutions — *trials, corrected ERSPC benefit, overdiagnosis, USPSTF D → C, MRI-first*
- Screening ↓ cancer death ~**20–30%**, building with time.

9 · Landmark trials

Trial	Design	Result	Take-home
ERSPC	>182,000; PSA every 2–4 y vs none	Cancer death ↓ 16% (9 y) → ~20% (16 y); excess incidence 85% → 36%	Benefit time-dependent
PLCO (US)	Annual PSA vs usual care	Control arm heavily screened	Contamination dilutes effect
Göteborg-1	Swedish screening	Significant ↓ cancer death	Supports ERSPC; basis of 4Kscore

Trial	Design	Result	Take-home
† **CAP** (UK)	Single PSA invitation	No benefit 10 y; tiny ↓ death 15 y	One-off screening weak
PCPT	18,882, PSA ≤3; finasteride 5 mg 7 y	Cancer ↓ **24.8%** (18.4 vs 24.4%); Gleason 7–10 6.4 vs 5.1%; 18-y death ↓ 25% (NS)	Fewer cancers, more high-grade
REDUCE	PSA 2.5–10, prior negative biopsy; dutasteride 0.5 mg 4 y	Cancer ↓ **22.8%**; Gleason 8–10 0.9 vs 0.6% (years 3–4: 12 vs 1)	FDA warning
SELECT	35,533; selenium 200 µg ± vitamin E 400 IU	Stopped for futility; vitamin E ↑ risk (P = 0.06)	No supplements
VITAL	>25,000; vitamin D3 2000 IU 5 y	No prostate effect	Not preventive

154 Prostate Cancer Biomarkers

1 · Why biomarkers beyond PSA

- PSA is **organ-specific, not cancer-specific** → negative biopsies and overdiagnosis of GG1. Guidelines advise **secondary testing** (imaging or biomarker) before biopsy; target is **clinically significant cancer (GG ≥2)**. → *Ch 153*
- Settings: before initial/repeat biopsy; favourable-risk cancer (AS vs treatment); after RP (adjuvant/salvage).
- Generations: organ (PSA) → cancer (PCA3, TMPRSS2:ERG, kallikreins) → high-grade cancer (SChLAP1).

2 · PSA biology

- PSA = **hK3**, gene KLK3 on **19q13**; androgen-regulated serine protease that **liquefies semen** (cleaves semenogelin). Half-life **2–3 days**.
- **proPSA** (244 aa, 7-aa leader) → **hK2 cleaves leader** → active PSA (237 aa).
- Serum rise = leak through lost basal layer/basement membrane. Per cell, cancer makes *less* PSA than benign tissue, but pattern 4 adds more serum PSA per cm^3.

Serum form	% of PSA	Measured?
PSA–α1-antichymotrypsin (ACT)	60–90	Yes (predominant, inactive)
PSA–α2-macroglobulin	10–20	**No** — epitopes masked
PSA–α1-protease inhibitor	1–5	Yes
Free PSA	5–45	Yes (proPSA, BPSA, intact PSA)

Confounders

- Biopsy: rise lasts ≥ **4 weeks**; cycling >55 km ~10%; ejaculation transient (24 h) — repeat after abstinence.
- **5ARIs**: PSA falls ~ **50%** by 12 months → **double** it (overestimates first 6 months, underestimates after years).
- † Age-specific upper limits (ng/mL): 40–49 → 2.5; 50–59 → 3.5; 60–69 → 4.5; 70–79 → 6.5.

3 · PSA derivatives

Derivative	Definition / values	Use
PSA density	PSA ÷ volume; cut-off **0.15** (0.08–0.15)	After negative/equivocal MRI; ≤0.15 = Johns Hopkins AS entry; predicts AS reclassification
PSA velocity	Change per year	Minimal value; **never an isolated biopsy trigger**
% free PSA	Cancer PSA binds ACT → **low %fPSA = cancer**; thresholds 15–25%	FDA: PSA 4–10, benign DRE; mainly **after negative biopsy**; unaffected by 5ARI

KEY POINT

%fPSA was validated against *any* cancer, not GG ≥2, so it overestimates significant risk. Risk calculators (PCPT, Prostate Biopsy Collaborative Group) include it.

- **[-2]proPSA**: peripheral zone, rises with Gleason → PHI. **BPSA**: transition zone, tracks BPH — no prognostic value. **Intact PSA**: in 4Kscore.

4 · Blood and urine tests (NCCN 2023 pre-biopsy)

Six tests estimate GG ≥2 risk; **thresholds are setting-specific** (higher for repeat biopsy).

Test	Components	Threshold / key fact
PHI	Blood: **(p2PSA ÷ fPSA) × √tPSA**	24–28.6 proposed; ~30% biopsies avoided
4Kscore	Blood: tPSA, fPSA, iPSA, **hK2** + age, DRE, prior biopsy	**7.5%**: sensitivity 94%, NPV 95%; ≤7.5% = low long-term metastasis risk
IsoPSA	Blood: PSA isoform structure	Unaffected by 5ARI or α-blockers
SelectMDx	Post-DRE urine DLX1, HOXC6 mRNA + clinical	NPV ~95%; needs prostate volume
ExoDx (EPI)	Urine exosomes: PCA3, ERG, SPDEF	**No DRE**; 15.6 initial, 20 repeat
MyProstateScore	Post-DRE urine PCA3 + TMPRSS2:ERG + PSA	10 initial (sensitivity 97%), 15 repeat; validated in Black men

- **PCA3**: long non-coding RNA (9q21–22). Score = (PCA3 ÷ KLK3 mRNA) × 1000 after **attentive DRE**; thresholds 25/35; **independent of prostate size**; FDA-approved after **previous negative biopsy**.
- **TMPRSS2:ERG**: **~100% specific** but in ~half of tumours → combined with PCA3.
- hK2 rises with cancer and grade independently of PSA.

> **KEY POINT**
> Urine tests follow a DRE because cancer RNA comes only from shed cells massaged into the urethra. ExoDx avoids this: exosomes are shed spontaneously. Urinary markers are highly specific; weakness is coverage → panels.

5 · Tissue-based biomarkers

- P2 Molecular staging of CA prostate — *Decipher, Oncotype GPS, Prolaris, PORTOS, ConfirmMDx; limits: heterogeneity, retrospective, G-MAJOR*
- **ConfirmMDx**: GSTP1, APC, RASSF1 methylation in negative cores; NPV ~88–90%; **field ("halo") effect**.
- Oncotype predicts primary pattern 4 or pT3 at RP. **Decipher ≥0.4 post-RP** may benefit from radiotherapy.

6 · Germline testing

- Up to 40–50% of risk familial; mostly **DNA-damage-repair** genes. STHLM3: 4 kallikreins + 232 SNPs + clinical variables.

Gene	Risk	Phenotype
BRCA2	4.5× <65 y; ~8.6× overall	Younger, higher grade, shorter survival
BRCA1	1.8× <65 y	—
ATM, CHEK2	CHEK2 OR ~2–3.4	ATM aggressive
Lynch (MSH2 etc.)	MSH2 ~10×	Also colorectal, upper tract
HOXB13 G84E	4.5×	**Not** more aggressive

NGS: BRCA1/2 alone or multigene panel (BRCA1/2, ATM, CHEK2, MLH1, MSH2, TP53, EPCAM). Indications → *Ch 153*.

7 · Clinical algorithm

ALGORITHM

Raised PSA → repeat (exclude infection, instrumentation, ejaculation, 5ARI) → MRI and/or biomarker (PHI, 4Kscore, IsoPSA, SelectMDx, EPI, MPS; PSA density) → biopsy if risk high

Negative biopsy, persistent suspicion → MRI, %fPSA, PCA3/EPI/MPS (repeat thresholds), ConfirmMDx → targeted re-biopsy

Favourable-risk → PSA density, Prolaris/Oncotype/Decipher → AS vs treatment

After RP → † PSA undetectable; ≥0.2 ng/mL confirmed = biochemical recurrence → Decipher/PORTOS for adjuvant/salvage RT

155 Prostate Biopsy: Techniques and Imaging

1 · Indications and contraindications

TRUS: PZ and central zone homogeneous echogenic, TZ heterogeneous. Most cancers arise in posterolateral PZ. Grey scale is poor: **39%** of cancers isoechoic; absence of a hypoechoic lesion never precludes biopsy. Tissue is mandatory.

- P2 TRUS / prostate biopsy — *indications, contraindications, preparation, technique, TR vs TP, complication rates*
- **Initial** also after cancer at TURP.
- **Repeat**: rising PSA; ASAP (**≈50%** cancer risk); atypical intraductal proliferation; positive PCA3/ConfirmMDx; suspicious MRI; recurrence after ablation; AS.
- NCCN: abnormal DRE with raised PSA, or repeat PSA >3 ng/mL with PI-RADS ≥3; PSA velocity alone, or abnormal DRE with normal PSA, insufficient. → *Ch 154*
- PSA density **>0.15 ng/mL/cc** suggests cancer. Volume = π/6 × transverse × AP × length; 1 cm^3 ≈ 1 g.
- **Contraindications**: coagulopathy, severe immunosuppression, acute UTI/prostatitis. Painful anorectal disease → anaesthesia. No anus → in-bore MRI or transperineal.

2 · Pre-biopsy mpMRI

- P1 P4 Prostate multiparametric MRI; PI-RADS and MRI–ultrasound fusion biopsy — *sequences, PI-RADS scoring, zone rules, cancer rate per score, targeting methods*

mpMRI (T2WI + DWI + DCE) is a **triage test**: no lesion → biopsy may be avoided; targeting finds more high-grade, fewer low-grade cancers. Lesion **>1.5 cm** or definite EPE → PI-RADS 5. AUA/SAR 2016: mpMRI and targeting of PI-RADS ≥3 after a negative biopsy with rising PSA.

Feature	Peripheral zone	Transition zone
Dominant sequence	**DWI**	**T2WI**
Dominant score 3	DCE positive → 4	DWI 5 → 4
† v2.1 (2019)	—	T2WI 2 with DWI ≥4 → 3

KEY POINT
DCE never creates or downgrades PI-RADS 4–5 and never upgrades a TZ lesion. mpMRI misses a minority of significant cancers (mostly 3+4): MRI-invisible lesions, reader discordance, registration/needle error, small lesion in large gland — so systematic cores remain relevant.

- Targeting: cognitive (position unrecorded), software fusion (UroNav electromagnetic; Artemis mechanical arm), in-bore. Fusion targeted cores ≈3× likelier than systematic to find intermediate/high-risk disease; PI-RADS 5 targets nearly always positive.
- **2–4 cores per target** (AUA/SUO 2023: 2–3, ± systematic). Indicated whenever biopsy is indicated and a PI-RADS ≥3 lesion exists; AS; focal-therapy planning.

3 · Systematic templates and technique

- **Sextant** (Hodge 1989) sampled too much TZ, missed lateral PZ. **Extended 12-core**: AUA standard. 6 → 12 raises detection, 12 → 18–21 does not: **geometry, not core number**.

- TZ/anterior cores: repeat setting or gland >50 mL. Seminal vesicle cores: palpable abnormality.
- 18-gauge gun (throw 15–25 mm; tip runs 0.5 cm beyond core): indent capsule, withdraw a few mm, fire. Never reposition with needle on rectal wall. ≤2 cores per jar.

4 · Transrectal vs transperineal (TP)

- TP: clean perineal skin (TR = Class III), no rectal bleeding, apex/anterior access (anterior-only cancers 6–20%). Templates: Ginsburg, MUSIC 12-core (omits midline → less pain/haematuria); none universal.
- EAU 2023: TP **preferred** (strong); detection non-inferior ("TREXIT"). Skin-flora cover or none — omitting antibiotics did not change the very low sepsis rate.
- Access: brachytherapy grid (5-mm mapping), freehand guide (PrecisionPoint), or two 14-gauge cannulas (cheapest; learning curve 4–6 weeks).

5 · Antibiotic prophylaxis

- Transrectal: every patient. AUA 2019: fluoroquinolone; 1st/2nd-generation cephalosporin ± aminoglycoside; or 3rd-generation cephalosporin. ≥1 h before; duration **≤24 h**.
- Single doses: levofloxacin 500 mg, gentamicin 5 mg/kg, amikacin 15 mg/kg, ceftriaxone 1–2 g. Renal impairment: aztreonam replaces aminoglycoside.
- FDA 2016 fluoroquinolone warning: tendon rupture, neuropathy, CNS effects, myasthenia.
- Infection risk: antibiotics within **6 months** (most consistent), diabetes, immunosuppression, prior UTI/biopsy infection, many prior biopsies, travel (India, South-East Asia), health-care worker, rectal resistant *E. coli* → augmented or swab-targeted prophylaxis, or TP.
- Endocarditis: IV ampicillin (vancomycin if allergic) + gentamicin → oral quinolone 2–3 days. Enema: no proven infection benefit.

6 · Anticoagulation and anaesthesia

Agent	Action
Low-dose aspirin	Continue
Warfarin, clopidogrel	Stop 5–7 days; biopsy when INR <1.5
DOACs	Stop 2–5 days
Mechanical valve	Bridge with heparin

- Transrectal: periprostatic block, 1–2% lidocaine 5 mL each side at SV–prostate junction (beats intrarectal gel).
- Transperineal: skin 1–2 mL → **periapical triangle** 5 mL (levator ani lateral, rhabdosphincter medial, external anal sphincter posterior, apex superior) → 3 mL along capsule.

7 · Complications and management

Complication	Rate	Management
Haematuria	23–63%	Clot retention (0.7%) → catheter, washout
Haematospermia	10–50%	Lasts ≈1 month
Rectal bleeding	2–22%	Ladder (below)
Infection / UTI	0.1–7% / 2–6%	Oral or IV antibiotics

Complication	Rate	Management
Sepsis	**0.3–3.1%**	Carbapenem until cultures
Vasovagal	1.4–5.3%	Trendelenburg, fluids
Retention	0.2–0.4%	Temporary catheter

ALGORITHM

Post-biopsy fever → assess sepsis → cultures → **no empirical fluoroquinolone** (resistant *E. coli*, also to co-trimoxazole, ampicillin) → carbapenem (or amikacin/3rd-generation cephalosporin per sensitivity) → resuscitate.

ALGORITHM

Persistent rectal bleeding → pressure → tamponade (sheath or Foley balloon) → endoscopy with adrenaline/sclerosant → angioembolisation → transrectal suture.

KEY POINT

Commonest complication: bleeding overall; infection among serious ones/admissions (1.0% → 4.1% in the resistance era). Frequent-trivial vs rare-dangerous → universal prophylaxis, move to perineum.

8 · Repeat and saturation biopsy

- P2 Saturation biopsy in CA prostate — *>12 cores, template mapping, indications, no gain at initial biopsy*
- Yield falls each round: ERSPC **22% → 10% → 5% → 4%** on biopsies 2–4. Third/fourth only for high suspicion; now mpMRI ± targeted, ConfirmMDx, anterior/TZ cores, TP route. Transurethral biopsy obsolete.

9 · Landmark trials

Trial	Design	Result	Take-home
PROMIS (2017)	576: mpMRI, TRUS, 5-mm template mapping (reference)	Sensitivity 93% vs 48%; NPV 89% vs 74%; 27% avoid biopsy	mpMRI better triage
PRECISION (2018)	500 biopsy-naive: MRI → targeted vs 12-core TRUS	Significant cancer 38% vs 26%; 28% avoided biopsy	MRI pathway superior
† MRI-FIRST (2019)	Paired systematic + targeted	Each alone missed cancers	Combine for maximal detection
PREVENT (2024)†	TP no antibiotics vs transrectal targeted prophylaxis	Infection not significantly different; detection comparable	Both acceptable

10 · MCQ one-liners

MCQ ONE-LINERS

1. Commonest false-positive hypoechoic lesion: BPH foci and prostatitis.
2. Significant cancer in PROMIS: Grade Group ≥3 or core length ≥6 mm.
3. Absent seminal vesicle: ipsilateral renal agenesis in 79%.
4. Müllerian cysts midline; wolffian cysts lateral.

156 Pathology of Prostatic Neoplasia

1 · Precursor and intraductal lesions

- P2 PIN — *definition, histology, precursor evidence, LGPIN not reported, isolated HGPIN needs no repeat biopsy, cribriform PIN reclassified*
- ~**20%** of HGPIN carries **TMPRSS2:ERG**. No repeat biopsy for isolated HGPIN (unifocal or multifocal, or on TUR) once mpMRI, DRE and PSA are considered.
- **IDC-P**: atypia beyond HGPIN (dense cribriform), **basal cells retained**; usually **retrograde spread of high-grade invasive cancer**. Prevalence ~2% low risk → >50% metastatic. Independently predicts early BCR. GG1 + IDC-P → upgraded at RP in **>80%**. Strict (Guo–Epstein) criteria license definitive therapy without proven invasion.

Lesion	Defining feature	Action on isolated finding
LGPIN	Mild atypia; irreproducible	Not reported
HGPIN	Benign architecture, atypical cells; non-cribriform	No repeat biopsy
PINATYP	HGPIN + adjacent small atypical glands	Manage as atypical
AIP	Atypia beyond HGPIN, below IDC-P	Consider re-biopsy (if benign or GG1 elsewhere)
IDC-P	Dense cribriform, basal cells retained	Definitive therapy (or re-biopsy); **excludes AS**

KEY POINT

HGPIN is a weak risk marker; IDC-P usually marks high-grade cancer already present.
Traps: re-biopsy for HGPIN; surveillance for GG1 with IDC-P.

2 · Adenocarcinoma: location, spread, staging

- **Zones (McNeal)**: †PZ ~70% of gland and ~70% of cancers — main mass in T2 and **85%** of T1c; posterolateral, palpable. †TZ ~5% of gland, ~20% of cancers — BPH origin; anterior, impalpable, under-sampled, lower grade. †CZ ~25% of gland, ~5% of cancers — around ejaculatory ducts; rare but aggressive (SVI). Periurethral glands <1% (median lobe).
- **Anterior tumours** ~15% of RPs (EPE more often margin-positive). **Multifocal >85%**.
- No true capsule → "**extraprostatic extension**". Exit via **perineural spaces**, posterolaterally.
- **SVI** = tumour in SV *muscular wall*; commonest route: out at base → periseminal tissue → SV; less often along ejaculatory ducts.
- Metastases: nodes, bone; lung at autopsy.
- EPE uncommon <**0.5 cm³**; nodes/SVI uncommon <**4 cm³**.

Stage	Definition
T1a / T1b	Incidental at TURP/enucleation; ≤5% / >5% of tissue
T1c	Non-palpable, needle biopsy
pT2	Organ-confined at RP
pT2+ (pT2x)	Positive margin at intraprostatic incision or ill-defined boundary (apex)
pT3a	EPE; **includes microscopic bladder neck invasion**
pT3b	SVI

Stage	Definition
pT4	†Other adjacent structures (rectum, levators, external sphincter, pelvic wall)

Trap: microscopic bladder neck = pT3a; T1a/b hinge on 5%, not grade.

3 · Grading: Gleason and grade groups

- P2 Gleason's score and its significance — *history, scoring rules, significance, limitations*
- **Biopsy (2005) = commonest + highest**; single pattern doubled. Lowest score in practice **6**; modern 6 has **no nodal metastatic capacity**, EPE uncommon.

Pattern	Morphology
3	Discrete well-formed glands
4	**Cribriform (incl. glomeruloid)**, poorly formed, fused
5	Sheets, cords, single cells, solid nests, **comedonecrosis**

GG (ISUP; WHO 2016)	Gleason	Nodes at RP	5-yr BCR-free
1	3+3	0%	**96%**
2	3+4	0.6%	**88%**
3	4+3	4.3%	**63%**
4	4+4, 3+5, 5+3	—	**48%**
5	9–10	—	**26%**

KEY POINT

Steep drop between GG2 and GG3 — never lump "7". 3+5 and 5+3 are GG4; GG5 starts at 9. Pure GG1 cannot metastasise, so calling it "cancer" is debated.

- **% pattern 4** in score 7 → report for GG2–3† (ISUP 2014). GG2 with limited pattern 4 may suit AS; GG3 with 90% pattern 4 → neoadjuvant hormones with RT; at 60% possibly avoidable.
- **Tertiary pattern 5** (<5% at RP): raises BCR risk but **stays in its GG** (nodal risk not raised).

Issue	ISUP 2019	GUPS 2019
% pattern 4 in GG2–3; cribriform	†Report	†Report
IDC-P with invasive cancer	†**Incorporated into grade**	†**Not graded**; comment
Targeted cores of one lesion	†**Aggregate grade** (plus per-core)	†Each core; aggregate optional

- †Biopsy: any higher pattern counts as secondary however small; RP: higher pattern <5% = tertiary, GG unchanged. Cribriform = pattern 4 regardless of extent.

KEY POINT

†Cribriform and IDC-P mark aggressive disease and argue against AS even in GG2, so they must be reported. Aggregate grading of a targeted lesion avoids under-grading (one core) and over-grading (highest core).

Upgrading setting	Risk
Biopsy GG1 → RP	~⅓
Biopsy GG2 → RP	~20% become GG3
GG1 with negative mpMRI (PI-RADS 1–2)	Still ~20% at 6 yr

Predictors: more biopsy extent, higher PSA, smaller gland, fewer cores. Reclassification <3 yr = missed, not progression.

4 · Needle biopsy assessment

- **Separate jars per site**; ≤2–3 cores/block. Pitfalls: **base — SV/central zone mimic HGPIN; apex — Cowper gland mimics cancer**.
- Report **positive cores + one other extent measure**; none superior.
- **Perineural invasion**: report on biopsy (↑ EPE; adverse on AS; prognostic for EBRT); omit at RP.
- **HMWCK, p63** stain basal cells; **absence supports cancer** (pitfalls: atrophy, adenosis). **AMACR**: adjunct only.
- **Adenosis** — chief mimic of low-grade cancer; TZ; **no increased cancer risk**.
- **ASAP**: ~5% of biopsies; cancer ~**40%**, PSA does not stratify; ~80% GG1, ~5% GG4–5. **Expert review before re-biopsy**, then re-biopsy sampling the atypical site.

> **KEY POINT**
> Sampling error is one-directional: a bad biopsy is believed, a good one is provisional, an atypical one is re-examined before it is repeated.

5 · TUR and RP specimens

- TUR: 6–8 cassettes detect all T1b. **Grade each major nodule**.
- **Nodes**: uncommon with Gleason <8. Node-positive: 15-yr BCR-free ~7%, CSS ~58%. Intraductal/cribriform → ↑ metastasis.
- **EPE**: focal vs nonfocal predicts progression. **SVI**: **65%** 5-yr progression.
- **Margins**: ~15% positive (robotic); only ~50% progress. Distance to ink irrelevant. Report **extent (focal vs nonfocal, 3 mm) and grade at margin**.
- **Intraprostatic incision**: risk equals focal EPE with positive margin.
- Independent predictors of BCR: grade, EPE extent, margins, SVI, nodes. **Tumour volume is not independent**.

6 · Treatment effect

Treatment	Histological effect	Gleason score?
Hormone therapy	Grade artefactually higher	**No** where effect present
Finasteride	No consistent change	Yes
Radiotherapy	Radiation atypia (up to 72 months)	Only if no effect
Cryotherapy / HIFU	Infarction; residual cancer unaltered	**Yes**

Post-RT biopsy: cancer with treatment effect has prognosis ≈ no cancer.

7 · Subtypes and other tumours

Subtype	Grading	Behaviour
Ductal (pure 0.4–0.8%; mixed ~5%)	**Pattern 4** (PIN-like 3; comedonecrosis 5)	Exophytic at verumontanum, mimics papillary urothelial cancer; PSA may be normal; ≈ Gleason 8; **visceral recurrence at low PSA**
Mucinous	As usual	As acinar

Subtype	Grading	Behaviour
Neuroendocrine differentiation (~half of acinar)	As usual	Not prognostic
Small cell (~50% mixed)	**Not graded**	Survival <1 yr
Squamous	—	**Osteolytic** metastases; hormone-unresponsive
Sarcomatoid	—	Dismal

- **Urothelial carcinoma in prostate** (35–45% of cystoprostatectomies): intraductal only → bladder stage decides; from urethral CIS: suburethral pT1, stromal pT2; **transmural from bladder = pT4**.
- IHC: prostate — PSA, P501S, NKX3.1; urothelial — **GATA3**.
- Sarcomas: **rhabdomyosarcoma in children, leiomyosarcoma in adults**. STUMP: may recur → follow-up.

157 Diagnosis and Staging of Prostate Cancer

1 · Clinical presentation

- Early cancer is **asymptomatic**; symptoms imply advanced disease.
- **Locally advanced**: voiding obstruction, ureteric obstruction (loin pain, renal failure), haematuria, reduced ejaculate.
- **Metastatic**: bone pain, fracture, leg oedema, cord compression; rarely retroperitoneal fibrosis, paraneoplastic syndromes, DIC.

2 · Screening in brief

- P2 PSA screening for prostate cancer: evidence, problems and solutions — *ERSPC vs PLCO, NNI 570 / NND 18, USPSTF D → C, AUA, NCCN, overdiagnosis*

3 · DRE and PSA

- **DRE**: low screening value at PSA <3, but abnormal DRE raises risk of GG4–5 and multiplies risk at every PSA level; DRE barely changes PSA.
- **PSA**: organ-specific, not cancer-specific; ↑ ≈4% per mL volume; serum PSA without cancer comes mainly from the TZ.
- Lower PSA: hypogonadism, obesity, statins, 5-ARIs, BPH surgery. On 5-ARI (finasteride 1 or 5 mg, dutasteride): **double PSA** after ≥12 months, or treat the nadir as new baseline.
- Risk is **continuous**: PCPT found cancer in 15% with PSA ≤4 (≈15% of these Gleason ≥7). Thresholds 4, 2.5–3 (ERSPC 3); 10 suggested >70 y. Repeat raised PSA before biopsy; no empirical antibiotics without urinary symptoms.

PSA (ng/mL)	Yield, normal DRE	Yield, suspicious DRE	High-grade (normal / suspicious)
0–1	8.8%	—	0.9%
4–10	17–32%	45–51%	4.1% / 11.7%
>10	43–65%	70–90%	19.4% / 50.5%

- Baseline PSA in the 40s predicts metastasis and death >25 years later (Malmö).
- PSA density ≥0.15; velocity >0.75 ng/mL/yr; free PSA ≤25% (PSA 4–10). → *Ch 154, 155*

4 · Staging: TNM (AJCC 8th, 2017)

Pathological stage (RP specimen, nodes) is more accurate and prognostic than clinical stage (DRE, PSA, biopsy, imaging).

Category	Definition
T1a / T1b	Incidental at TUR: ≤5% / >5% of tissue
T1c	Impalpable, found on needle biopsy
T2a / T2b / T2c	Palpable, confined: ≤½ one side / >½ one side / both
T3a / T3b	Extraprostatic extension / seminal vesicle invasion
T4	Fixed or invading sphincter, rectum, bladder, levators, pelvic wall
N1	Regional (pelvic) nodes
M1a / M1b / M1c	Non-regional nodes / bone / other ± bone

AJCC prognostic groups add PSA and GG. Whitmore–Jewett is historical.

> **KEY POINT**
> T1 is defined by how cancer was **found**, T2 by what is **felt**. Traps: SVI is T3b; non-regional nodes are M1a; biopsy-found impalpable cancer is T1c even if bilateral; clinical T is by DRE, not MRI.

5 · Predicting extent and risk

- **PSA** cannot stage an individual. Organ-confined ≈80% at PSA <4, 66% at 4–10, <50% >10. Nodes ≈20% at >20, **75% at >50**.
- **Biopsy grade** is the most important biopsy finding. → *Ch 156* Tertiary 5 on biopsy becomes secondary (3+4 + tertiary 5 → 3+5). Tumour in SV or periprostatic fat on biopsy worsens prognosis. Acid phosphatase abandoned.

D'Amico	Stage	PSA	Gleason	10-yr disease-free after RP
Low (all)	T1–T2a	≤10	≤6	83%
Intermediate (any)	T2b	>10–20	7	46%
High (any)	T2c	>20	8–10	29%

† NCCN group	Criteria
Very low	T1c, GG1, PSA <10, PSAD <0.15, <3 positive cores, ≤50% of any core
Low	T1–T2a, GG1, PSA <10
Favourable intermediate	One intermediate factor (T2b–c, GG2–3, PSA 10–20), GG1–2, <50% cores positive
Unfavourable intermediate	≥2 intermediate factors, or GG3, or ≥50% cores positive
High	T3a, or GG4–5, or PSA >20
Very high	T3b–T4, primary pattern 5, >4 cores GG4–5, or ≥2 high-risk features

- **CAPRA** score; **nomograms** beat groups for individuals (†Partin tables, MSKCC); nodal nomograms decide PLND.
- **Tissue genomic tests** (Prolaris, Oncotype DX 17 genes, Decipher 22 RNA) refine risk; prospective validation awaited.

6 · Staging imaging and nodes

Risk	Imaging
Low / intermediate, asymptomatic	No routine CT or bone scan
High	Bone scan + CT or mpMRI; consider PSMA PET

- **mpMRI**: local staging (EPE, SVI), planning, AS selection.
- **PSMA PET**: ^{68}Ga-PSMA-11 and piflufolastat F-18 FDA-approved for high-risk staging; more sensitive/specific for nodes and metastases; benefit when conventional imaging negative unproven.
- **PLND**: no consistent oncological benefit — value is **staging**; perform when nomogram nodal risk exceeds guideline threshold.

KEY POINT

Imaging yield follows pre-test probability: in low-risk men most positive scans are false positives → more tests, anxiety, wrong denial of cure.

7 · Landmark trials

Trial	Design	Result	Take-home
ERSPC	European screening RCT, from 1993	**20% relative reduction** in cancer death at 16 y	Saves lives, at cost of overdiagnosis
PLCO	US screening RCT, from 1993	No mortality difference	Contaminated control; ≈3× fewer events
PCPT	Placebo arm biopsied regardless of PSA	15% cancer at PSA ≤4	No safe PSA level
† proPSMA (2020)	PSMA PET-CT vs CT + bone scan, high-risk	Accuracy ≈92% vs 65%; less radiation	PSMA PET better first-line staging

8 · MCQ one-liners

MCQ ONE-LINERS

1. Finasteride 1 mg lowers PSA as much as 5 mg.

158 Active Management Strategies for Localised Prostate Cancer

1 · The modern question: timing as well as modality

- 15-year randomised data in favourable risk: surveillance → **more metastasis but no difference in cancer mortality**.
- **Spectrum**: competing risks dominate → **watchful waiting (WW)**; good longevity, benefit unclear → **active surveillance (AS)**; needs therapy → RP, EBRT, brachytherapy, or whole/partial-gland ablation.

> **KEY POINT**
> Cancer mortality is always weighed against competing mortality. A competing risk argues against treatment only when it will act *before* the cancer does.

2 · Host evaluation

- NCCN: estimate life expectancy (SSA or WHO life tables). PCOS: 10-year prostate cancer mortality **3–18%** by risk vs **≥33%** other-cause with any comorbidity.
- The blunt **"10-year life expectancy" rule** undertreats older and overtreats younger men → individualise. **Age bias**: clinicians judge by age, not comorbidity.
- **Charlson (CCI)**: designed for 1-year inpatient mortality; **PCCI** reweights for 10-year. **CAN score** (automated, VHA): best predictor of 5-year survival MCQ.
- NNT rises steeply with Charlson score.

Baseline function (IPSS, UCLA Prostate Cancer Index)

Baseline state	Favours	Reason
Obstructive LUTS, large gland	Surgery	RP relieves obstruction; brachytherapy ↑ retention
Severe irritative LUTS	Surgery	Worst after radiation
Gland >60 cm³	Surgery (or neoadjuvant ADT)	Pubic arch interference; >80 g → ADT (shrinks 25–40%) before RT
Chronic retention / neurogenic bladder	Surgery	RT → bladder fibrosis; if RT chosen, switch to intermittent self-catheterisation
Prior TUR defect	Surgery	Brachytherapy dosimetry hard; ↑ incontinence. TURP *before* RT, delay EBRT **6–8 weeks**
IBD, bowel dysfunction, prior rectal surgery	Surgery	RT bowel toxicity

- **Sexual** recovery depends on age, baseline erection, PDE5 inhibitor use, vascular risk; poor baseline → more non-nerve-sparing surgery.

Modality	Absolute contraindication	Relative
Radiotherapy	Previous radical pelvic irradiation	Bowel in field (prior surgery, hernia); uncontrolled diabetes; IBD; bowel dysfunction
Surgery	—	Previous pelvic RT; chronic steroids; severe cardiac/pulmonary/haematological comorbidity

3 · Risk stratification

NCCN group	Criteria	Staging imaging
Very low	**All**: T1c; GG1; PSA <10; <3 positive cores, ≤50% each; PSA density < **0.15**	Not indicated
Low	**All**: T1–T2a; GG1; PSA <10	
Favourable intermediate	IRFs: T2b–c, GG2, PSA 10–20. **All**: 1 IRF; GG1–2; <50% cores positive	
Unfavourable intermediate	**Any**: 2–3 IRFs; GG3; ≥50% cores positive	Bone scan + CT/MRI, chest imaging; PSMA PET may replace (NCCN 2023)
High	**Any**: T3a; GG4–5; PSA >20	
Very high	**Any**: T3b–T4; primary pattern 5; >4 cores GG4–5	

KEY POINT

Favourable groups need ALL criteria; adverse groups ANY one. PSA density appears only in very low risk; GG3 alone makes intermediate unfavourable. AUA/ASTRO/SUO use very low/low/intermediate/high without substratification.

- **Epstein criteria** (insignificant cancer): PSA density <0.15, Gleason ≤6, <3 positive cores, ≤50% any core.
- Nomograms: **Partin** (pathological stage, accurate 72.4%); **Kattan** (relapse after RP); **UCSF-CAPRA** (risk jumps at score ≥6).
- **proPSMA** (302 high-risk): accuracy **92% vs 65%**, sensitivity 85% vs 38%; negative scan cannot exclude small nodes.
- Genomic tests: AUA 2022 — not routine; only where they change an AS decision (high-volume 3+3, or 3+4 considering AS).

4 · Observation vs active surveillance

- P2 Expectant treatment of CA prostate — *AS (curative, deferred) vs WW (palliative): candidate, monitoring, triggers*
- AS **preferred for very low and most low risk** (AUA, ASTRO, SUO, NCCN); selected low-volume GG2 acceptable.
- Limitations: **25–45%** occult high grade on systematic biopsy; declining adherence; no agreed triggers. Reclassification predictors: PSA density >0.15, core cancer density. *→ Ch 159*

5 · Comparative oncological evidence

- Biochemical-free survival is a flawed RP-vs-RT end point. Population studies (Swedish SHR ≈1.8; PCOS HR 0.35 disease-specific, curves apart by 3 years) favour RP — largely **selection bias**.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RP vs WW; 695; pre-PSA, ~90% palpable	23.2 yr: cancer deaths 63 vs 99, RR **0.56**; absolute 11%; NNT **8**; most benefit <65 (RR 0.45), intermediate risk	Benefit in younger, intermediate risk

Trial	Design	Result	Take-home
PIVOT	RP vs observation; 731 (planned 2000); PSA era; older, comorbid	10 yr: no difference; benefit if PSA >10. 19.5 yr: all-cause ARR 5.5% (NS), cancer 4.0% (NS); significant only in **intermediate risk** (ARR 14.5%). Bone metastases 4.7% vs 10.6%	No benefit in low risk; underpowered
ProtecT	1643 screen-detected, 50–69; monitoring vs RP vs RT + 3–6 mo ADT; 77% low risk	15 yr cancer mortality **2.7% vs 2.2% vs 2.9%** (P = 0.53); more metastasis with monitoring; 60% of monitored treated by 15 yr	AS safe in favourable risk; RP = RT

KEY POINT

Trials show little benefit because competing mortality and low-risk enrolment dilute the effect; benefit appears only in younger intermediate-risk men with long follow-up. ProtecT expected 15% mortality, saw ~1% at 10 years.

7 · Functional outcomes compared

Domain (vs AS; Chen, JAMA 2017)	RP	EBRT	Brachytherapy
Sexual	**Worst**; persists at 24 mo	Mild early decline; near AS by 24 mo	Mild early decline
Incontinence	**Worst**; persists at 24 mo	No change	No change
Obstructive/irritative	Worse at 3 mo, then **better than AS**	Worse at 3 mo; resolves by 12	**Worst acute**; resolves by 24 mo
Bowel	No change	**Worse**, persists at 24 mo	No change

- **PCOS**: differences at 2 and 5 y; **none at 15 y**.
- **ProtecT**: RT rising bloody stools; no difference in anxiety, depression or QoL.
- **CEASAR**: RP erectile disadvantage confined to top quartile of baseline function.
- Open vs robotic RP: one RCT — similar function at 12 weeks; surgeon matters, not approach.
  → Ch 160/161

KEY POINT

Surgery front-loads urinary and sexual harm that partly recovers; radiation spares early function but adds bowel toxicity and slow decline. Curves converge by 15 years — the choice is *when and where* the patient accepts harm.

8 · Shared decision-making

- No treatment clearly superior → **patient preference carries unusual weight** (AUA, ASTRO, SUO). The **urologist's recommendation is the most influential factor** — beware bias. **Longevity assessment may be the most important counselling component**.

9 · Stepwise decision algorithm

ALGORITHM

1 — Treat or not? Disease (grade, stage, genomics, family history) + host (age, comorbidity, longevity, preference) → WW (short longevity) or AS (very low/low, selected favourable intermediate).

2 — Partial or whole gland? Focality, size, grade, potency, LUTS → focal ablation (limited-focality intermediate risk, ideally in a trial) *→ Ch 163*.

3 — Surgery vs radiation vs whole-gland ablation → outcomes, toxicities, baseline function, secondary treatment risk.

4 — Delivery: open vs robotic RP, node dissection, surgeon; EBRT vs brachytherapy, fractionation, ADT *→ Ch 162*.

Risk / longevity	Usual recommendation
Limited longevity, low lethal potential	WW; ADT at symptoms
Very low	AS (best option)
Low	AS preferred for most; all therapies options
Favourable intermediate	RP or RT; AS for select (low-volume GG2); focal therapy in trials
Unfavourable intermediate	RP or RT + ADT; AS strongly discouraged
High / very high	RP or RT + ADT; consider even with competing risks if cancer threatens within 10 yr; no focal therapy

- **Surgery**: early incontinence/ED are the norm — set realistic timeline; quote surgeon-specific outcomes; LUTS may improve; age <65 lower cancer mortality (ProtecT).
- **Radiation**: spares short-term function; yearly decline; secondary rectal/bladder cancer at 15–20 y (controversial); ADT for intermediate/high risk drives early toxicity.
- **Whole-gland ablation**: little long-term data; salvage surgery hampered by fibrosis. Cryoablation → ED almost always; HIFU → better function but retreatment **29–38%**.
- EBRT, brachytherapy, SBRT, protons: comparable 10-year oncology.

10 · MCQ one-liners

MCQ ONE-LINERS

1. Early divergence of RP vs RT survival curves in observational studies indicates selection bias.
2. Brachytherapy, not EBRT, causes the worst acute obstructive/irritative symptoms.

159 Active Surveillance of Prostate Cancer

1 · Rationale: overdiagnosis

- **Latent reservoir**: autopsy cancer ≈ age as %; PSA + biopsy detects 15–25%, but lifetime risk of cancer death is only **3%**. Harbouring : death ≈ 20:1; diagnosis : death ≈ 6:1; overdiagnosis 87–94%.
- **Two premises**: long lead time to progression, and cure still possible at first sign of higher risk.

2 · AS versus watchful waiting

- P2 Expectant treatment of CA prostate — *AS curative/deferred vs WW palliative; candidates, monitoring, switch at life expectancy <5–10 yr*
- **Pitfall — neuroendocrine dedifferentiation**: progression without PSA rise in elderly men with unrecognised hypogonadism → measure testosterone, interval imaging.

> **KEY POINT**
> AS defers curative intent; WW abandons it. Life expectancy, not grade, chooses between them.

3 · Biology of grade group 1

- **Pattern 3 is molecularly near-normal**; patterns 4–5 show EGFR overexpression, HER2 amplification, BCL2, angiogenesis, **PTEN deleted in 70–90%**; cyclin D2 is lost in pattern 4. **MYC amplification + PTEN deletion** drives metastasis.
- **Metastatic potential of pure GG1 effectively zero**; occasional EPE.
- True grade progression ~**1–2% per year**, mostly high-volume Gleason 6 → long-term follow-up.
- Deaths in low-risk series = **occult higher grade**, present in ~**30%** after systematic biopsy (85% of it 3+4).

Hazard	Misattribution of grade	True progression
Frequency	**Commonest (25–30%)**	1–2.3%/year
Usual result	Mostly GG2	Many 3+4 stay indolent
Remedy	MRI, targeted biopsy, predictors	Long-term surveillance

> **KEY POINT**
> The main hazard of AS is **sampling error at entry**, not biology over time ("wolf in sheep's clothing"). First job: get the entry grade right.

4 · Patient selection

Most GG1 should be on AS (AUA, EAU, ASCO, NICE, Cancer Care Ontario).

	Very low risk (Epstein)	Low risk
Definition	GG1; **≤2 positive cores; no core >50%; PSA density <0.15**	Any GG1, PSA <10
Progression to treatment	Lower	Higher

	Very low risk (Epstein)	Low risk
Metastasis	Exceptionally low	Exceptionally low

The split reflects likelihood of **occult higher grade**. **High-volume GG1**: candidate, but scrutinise with mpMRI + targeted biopsy, re-biopsy and/or genomic assay.

- **Toronto** (inclusive: all Gleason 6 + selected intermediate; ~50% eligible) 15-yr cancer mortality **5%** actuarial (1.5% actual) vs **Johns Hopkins** (Epstein very low; ~20% eligible) **0.5%**. **Inclusion criteria set mortality; intervention constant (~25% at 5 yr)**. Toronto baseline Gleason 7: ≥20% metastasis at 15 yr (HR 3.75); PSA >10 barely predictive.
- PRIAS (n=5302): cancer mortality <1% at 10 yr. Other-cause mortality far exceeds cancer mortality; cardiovascular disease commonest cause of death.
- **Favourable intermediate**: AS selectively for small-volume GG2. Only NICE lists AS and radical treatment as equal for intermediate risk. Exclude GG ≥3; intraductal/cribriform raises concern.
- **Black men**: inconsistent data; AS candidates, perhaps more intense surveillance.
- **Family history** rarely justifies treatment ("family history inflation"); exception: metastatic disease <60 yr in first-degree relative → germline testing.
- **BRCA2**: highest upgrading risk (to GG ≥3 within 10 yr); BRCA1, ATM less. Consider definitive therapy even for low grade; AS only with very close follow-up.
- Higher-grade risk factors: **PSA density >0.15**, Black race, biopsy volume; restricted diffusion on MRI powerful.

5 · Monitoring protocol

ALGORITHM

Diagnosis → mpMRI → **confirmation biopsy** (systematic + targeted) within 1–3 yr → PSA 6-monthly, DRE every 1–2 yr, MRI ≤ every 18–24 months → repeat biopsy every 3–5 yr while stable → trigger → definitive therapy, or WW when life expectancy falls

- Confirmation biopsy may be omitted if initial biopsy MRI-targeted; delay to year 3 harmless (modelling). **Compliance is paramount**: compliers had better cancer mortality and metastasis-free survival.

Guideline	Intermediate risk	Schedule / MRI
Cancer Care Ontario	Treat; AS if selected	PSA 3–6-monthly; DRE yearly; biopsy within 6–12 mo then 3–5-yearly; MRI if discordant
AUA/ASTRO 2022	AS if selected	PSA 6–12-monthly; DRE 1–2-yearly; biopsy 1–4-yearly; MRI before initial or confirmation biopsy
NICE 2019	Radical treatment or AS	PSA 3–4-monthly year 1, then 6-monthly; **MRI at enrolment**

Prompts definitive therapy	Prompts evaluation only
GG ≥3 (unless life expectancy significantly shortened)	More Gleason 6 volume → MRI or biomarker
Traditionally any Gleason ≥7 (now relaxed)	Small-volume GG2 → may continue AS
	PSA rise alone → repeat MRI or biopsy

KEY POINT
Grade triggers treatment; volume and PSA trigger investigation. 10–15% leave AS by choice (anxiety) without a trigger.

6 · MRI in AS

- DWI signal inversely related to Gleason: MRI misses low grade but locates higher grade.
- **Entry — strong evidence**: mpMRI before confirmation biopsy with **systematic + targeted cores** (~⅓ upgraded, found by both). NICE recommends; NCCN option.
- **Serial monitoring alone — weak** (NPV ~0.8–0.9 for GG ≥2) → never the sole means of excluding progression. MRI-invisible cancers behave better; visible lesion predicts treatment. Higher-risk men still need systematic biopsy.

7 · Biomarkers and genomics

- Decipher (22-marker RNA): predicts upgrading in GG1, not GG2. Oncotype DX: mixed, none in Canary PASS. Prolaris: no AS-specific evidence. None built for AS; gain modest.
- **Bayesian problem**: with 3% prior metastasis risk, a 90%-specific test labels 10% "significant" — false positives ≈ 3× true risk → test only higher-risk candidates (high volume, PSA density, GG2).
- Set surveillance intensity; **never sole basis for AS**; favourable score does not override adverse clinicopathology.

8 · Lifestyle, drugs and psychology

- **Stop smoking**, vigorous exercise (ERASE: lowered PSA), healthy weight, vegetable diet.
- Only vitamin D 1000–2000 IU/day reasonable; **selenium and vitamin E contraindicated** (SELECT).
- 5ARIs not used to prevent progression (REDEEM). Statins reasonable for CVD; metformin, apalutamide, enzalutamide investigational.

9 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	695, pre-PSA, palpable; RP vs WW; 23 yr	Cancer death 19.6% vs 31.3% (RR 0.55)	Benefit age <65 and GG ≥3; none GG1–2
PIVOT	731 US veterans; RP vs observation; 18.6 yr	All-cause death 68% vs 73% (NS); low-risk cancer death 2.7%	Modest benefit, intermediate risk
ProtecT	1643 PSA-detected; monitoring vs RP vs RT; 15 yr	Cancer mortality <3%, no difference; **metastases 9.4% vs 4.5% vs 5.0%**; 61% of monitoring arm treated	Active monitoring ≠ AS (no scheduled biopsy, >⅓ intermediate/high)
ASIST	GG1 on AS; MRI-targeted + systematic vs systematic confirmation biopsy	Upgrading 33% vs 27% (NS); ~50% less AS failure at 2 yr	MRI benefit shows over follow-up
REDEEM	Dutasteride vs placebo on AS	38% less "progression" (less GG1 volume), no upgrading difference	Cytoreduction, not disease modification

160 Open Radical Prostatectomy

1 · Principles

- Radical prostatectomy (RP): only treatment shown in an RCT to reduce metastasis and cancer death (**SPCG-4**).
- Most RPs now robotic, without proven benefit → *Ch 161*.
- **Goals in order**: cancer control → avoid technical complications → continence → potency.

> **KEY POINT**
> The order of goals settles every intraoperative trade-off: when nerve sparing and a clear margin conflict, the margin wins.

2 · Surgical anatomy

- P1 P4 Anatomy of prostate and neurovascular bundles; Prostate anatomy, newer surgical concepts and NVB preservation — *fascial layers table, NVB of Walsh, inferior vesical urethral/capsular branches, accessory pudendals, Santorini plexus, nerve-sparing planes*
- DVC **common trunk lies directly over the urethra** — where it is transected.
- Urethral branches at the bladder neck, **5 and 7 o'clock** pedicles; capsular branches mark the cavernous nerves.
- Pelvic plexus = parasympathetic **S2–S4** + hypogastric sympathetic; beside the rectum, **5–11 cm** from the anal verge, midpoint at the seminal vesicle tip.
- At the apex NVB branches spray anteriorly and posteriorly → basis of **high anterior release**.
- **Striated sphincter** = vertical tubular sheath (Oelrich), not a flat diaphragm; apical dissection decides continence.
- Passive continence: sphincter (slow-twitch). Active: levator ani. Both mainly **pudendal** → pelvic floor exercises train both.

> **KEY POINT**
> Nerves and vessels run between levator and prostatic fascia. Leaving prostatic fascia on the gland and Denonvilliers fascia on its back lets nerve sparing and clear margins coexist.

3 · Selection and preoperative care

- **Harder, not contraindications**: prior pelvic surgery or radiotherapy, TURP, hernia mesh, large gland.
- **Defer 6 weeks after biopsy, 12 weeks after TURP**.
- Inguinal hernia ~15% preop, up to 20% after RP → examine standing; repair preperitoneally at RP.
- No autologous donation; cephalosporin prophylaxis.
- **Fibreoptic headlight** = most useful instrument.

4 · Retropubic RP — operative steps

> **ALGORITHM**
> Lower midline extraperitoneal incision → PLND → endopelvic fascia → puboprostatic ligaments → DVC → urethra → posterior sphincter → NVB release → pedicles → bladder neck and seminal vesicles → reconstruction → anastomosis

- **PLND**: intermediate/high risk; omitted for Grade Group 1–2 low-volume disease. *→ Ch 161*.
- **Endopelvic fascia**: open at sidewall reflection (too medial → Santorini bleeding).
- **Puboprostatic ligaments**: divide, but spare the **pubourethral** part (anchors sphincter).

DVC control

Feature	Stapler (preferred)	Suture ligation
Method	Vascular load; fire only if catheter moves freely	3-0 figure-of-eight to pubic perichondrium + back-bleeding stitch
Advantage	~200–400 mL less blood loss; less traction	Recreates puboprostatic support

- **Anterior apex = commonest positive margin site**. Absolute venous control before touching the apex: oversew distal DVC as a "hood".
- **Urethra**: transect close to apex; six 3-0 sutures, **mucosa and submucosa only**.

Nerve sparing

- **Interfascial** plane (levator vs prostatic fascia) is correct; Walsh never uses intrafascial (margin risk); wide (extrafascial) excision for induration, fixed bundle, doubtful margin.
- Both NVBs usually preserved: extracapsular extension rarely exceeds 1–2 mm; margin rates equal with or without sparing.
- Decided in theatre: **induration or fixed bundle → excise that side** (touch absent robotically).
- **Technique**: release prostate from bundle, not bundle from prostate; start **at the apex** → base; ligate posterior branch at seminal vesicle. Small clips, **never thermal energy**.
- **High anterior release** (before urethral division) → less traction, faster recovery; unilateral as good as bilateral, so favourable side only. Avoid if ≥2 **Tsuzuki criteria**: PSA >10, Gleason >6, >20% core involvement, >33% cores positive, perineural invasion, abnormal DRE.
- **Wide excision**: clamp **medial → lateral** on rectum (reverse risks rectal injury).

Pedicles, bladder neck, reconstruction

- **Complete seminal vesicle excision is essential**; pelvic plexus lies on their lateral surface. Doubtful posterolateral margin on specimen → excise that NVB.
- **Tennis-racquet** closure + mucosal eversion; **intussusception** sutures → "**turtle sign**".
- Tension-free anastomosis; peritoneal window lets haematoma decompress.

> **KEY POINT**
> Every step does one of three jobs: absolute venous control before the apex; the right fascia on the specimen; no traction or heat on the NVB.

5 · Perineal RP (brief)

- Reflects fasciae off the prostate, sparing the DVC → **less bleeding**; retropubic RP works outside the fasciae, so the DVC must be ligated.
- † PLND needs a separate approach.

6 · Intraoperative complications

Complication	Setting	Management
Haemorrhage (commonest, venous)	Endopelvic fascia, DVC	**Divide DVC completely and oversew** (best measure)
Obturator nerve	PLND	Reanastomose, fine non-absorbable
Rectal injury	Small prostate; apical/posterior plane	Two-layer repair + omentum; colostomy if prior radiotherapy
Ureteric injury	Dissecting in trigone	Reimplantation

- Typical loss 300–1000 mL; partial DVC division bleeds more.
- **Rectal repair**: finish prostatectomy → anal dilatation → 2-layer closure → omental pedicle between rectum and anastomosis (prevents rectourethral fistula).

7 · Postoperative care

- Discharge day 1–2. Ketorolac if creatinine <1.2 mg/dL, no haematuria.
- **Catheter ~1 week**; day-4 removal → frequent retention. Never remove with significant extravasation.
- No imaging needed if haematocrit fall <6%. Bladder spasm → oxybutynin 5 mg.

8 · Postoperative complications

- Mortality **0.2%**.
- **Delayed haemorrhage → early exploration and evacuation** (shortens stay).
- **Thromboembolism** = major cause of death; highest with PLND, peaks **14–28 days**. Compression, early ambulation; treat with LMWH.
- **Bladder neck contracture** (~3 months): dilatation → cold-knife incision 3, 6, 9 o'clock + self-catheterisation → triamcinolone → rarely reconstruction + artificial sphincter.
- **Lymphocele** → *Ch 161*.
- P2 Post-prostatectomy incontinence — *rhabdosphincter dependence, evaluation, PFMT, sling vs AUS*
- **Incontinence** = intrinsic sphincter deficiency (sphincter injury at DVC division). Prevent: preserve apical sphincter, precise tension-free mucosal anastomosis, small supple neck. Treat: less fluid, no caffeine, stop α-blockers, pelvic floor exercises. Continence = **0–1 pad/24 h**.
- **ED**: recovery predicted by **age <65, preoperative potency, no diabetes, bilateral nerve sparing**; continues up to 4 years. PDE5 rehabilitation helps; nightly not better than on-demand.

> **KEY POINT**
> Most late complications trace to the pelvic haematoma: it distracts the anastomosis (contracture, incontinence) and inflames the NVBs — hence early re-exploration, clips near the bundles and a peritoneal window.

9 · Outcomes

Time	Continent	Potent (Walsh)
3 months	80%	38%

Time	Continent	Potent (Walsh)
12 months	93%	73%
18–24 months	97%	86%

- Single-expert figures; another centre: only a third at baseline potency at 2 years.
- RP improves LUTS up to 15 years.

10 · Landmark trials

Trial	Design	Result	Take-home
SPCG-4	RCT, RP vs observation	Less metastasis and cancer death	RP cuts cancer death
Padma-Nathan	RCT, nightly sildenafil vs placebo	Better erections at 1 year	Rehabilitation works
REINVENT	RCT, vardenafil on-demand vs nightly	On-demand better	Daily dosing unproven

161 Laparoscopic and Robotic-Assisted Laparoscopic Radical Prostatectomy and Pelvic Lymphadenectomy

1 · Background

- P1 Open vs laparoscopic vs robotic radical prostatectomy; Robotic radical prostatectomy — *history, comparison table, contraindications, ports, steps, complication rates*

> **KEY POINT**
> The robot did not change the operation; it removed the suturing barrier. Surgical anatomy (dorsal venous complex [DVC], neurovascular bundle [NVB], fasciae, sphincter) is as in open surgery *→ Ch 160*.

2 · Patient selection

- Only absolute contraindications: uncorrectable bleeding diathesis, unfit for general anaesthesia; others add difficulty only.
- PSMA PET-CT if high risk with negative conventional imaging.
- Prior extraperitoneal mesh repair → **transperitoneal** route (Retzius obliterated).
- **Obesity**: longer trocars placed more cephalad, pneumoperitoneum 10 mm Hg. **Large gland/median lobe**: longer, more blood loss, continence comparable. **Prior TURP**: orifices close to bladder neck.
- mpMRI for extracapsular extension (ECE): specificity **0.88**, sensitivity **0.61**; guides extent of nerve sparing.

3 · Preparation and anaesthesia

- Consent for **conversion to open**; **general anaesthesia mandatory**.
- **Positioning**: arms tucked and padded; **no fixed shoulder rests** (brachial plexus); avoid exaggerated hip flexion (femoral neuropraxia).
- CO_2 pneumoperitoneum: vagal response, oliguria, hypercarbia → raise ventilation, lower pressure. Fluids restricted (1.5–2 L, mostly after anastomosis).
- **Steep Trendelenburg**: facial, laryngeal, corneal oedema, raised intraocular pressure → protect eyes; marked facial oedema → consider delayed extubation.

4 · Access

- **Extraperitoneal**: smaller space, bowel protected, less tilt (obesity, lung disease), more CO_2 absorption. **Transperitoneal**: larger space; preferred after extraperitoneal mesh. No consistent advantage either way.
- Transperitoneal **anterior** (commonest; Retzius first) vs **retrovesical** (seminal vesicles first — useful with a median lobe).
- Camera port **15–17 cm** from symphysis; 12–15 mm Hg.

5 · Operative steps (transperitoneal anterior RALRP)

ALGORITHM

Drop bladder (Retzius) → endopelvic fascia → DVC ligation → bladder neck → seminal vesicles and vasa → posterior plane (Denonvilliers) → pedicles and NVB release → apex: DVC and urethra → inspect → PLND → entrapment → posterior reconstruction → anastomosis → extraction

- Endopelvic fascia: sharp, **no cautery**. Preserve accessory pudendal arteries.
- **DVC**: pneumoperitoneum tamponades venous bleeding. Suture close to pubis, stapler, or delayed division and oversewing.

Bladder neck — no tactile feedback

- Cues: prevesical fat line, balloon traction, instrument "pinch".
- Divide anteriorly in midline → lift catheter → inspect for median lobe and orifices → divide posterior neck angled **45°** downward.
- Prior TURP: balloon falls into defect → division too distal. Balloon off midline → median lobe. When in doubt, **divide more proximally**.

Posterior plane, pedicles, nerve sparing

- P1 P4 Anatomy of prostate and NVB; NVB preservation in RP — *nerve-sparing planes, Denonvilliers excision for posterolateral ECE, injury mechanisms, apex = commonest margin*
- Incise Denonvilliers ~0.5 cm below vesicle bases. Bleeding = too close to prostate. Mobilise rectum widely early.
- **Intrafascial** accepted here in selected men (Ch 160 never uses it).
- **Athermal (veil) release**: incise levator fascia anteromedially → sweep NVB posterolaterally off white prostatic fascia → Hem-o-lok on pedicle → sharp division. Release NVB **before** mobilising specimen. NVB bleeding → fine sutures, not cautery.
- Pedicle clips near anastomosis can migrate into bladder → haematuria, infection, stones.

Apex and reconstruction

- Preserve urethral length but watch for an apical lip; a small urethral rim on the apex lowers margins without harming continence.
- Bladder neck reconstruction only for large gland, median lobe, prior TURP → tennis-racquet; protect orifices (methylene blue, stent).
- **Rocco stitch**: Denonvilliers remnant + posterior bladder → posterior rhabdosphincter. Continence benefit inconsistent; agreed benefit = **tension-free anastomosis**.
- **Van Velthoven running anastomosis**: from 6 o'clock, two arms run anteriorly (now barbed); no eversion sutures needed.

6 · Continence-preserving variants

- P4 Newer surgical concepts in CA prostate — *Retzius-sparing: earlier continence, more margins; Hood: early data*
- Retzius-sparing leaves detrusor apron, puboprostatic ligaments, DVC untouched; anastomosis anterior first ("upside-down"). Rationale: preserving anterior support beats reconstructing it.

7 · Pelvic lymph node dissection

- P4 P1 PLND templates; Lymphocele — *templates (extended ~63%), staging not survival; lymphocele drain → fenestration*
- Super-extended adds presacral nodes. Standard template misses most drainage → **if PLND is done, use the extended template**; morbidity rises with extent.
- **Indications**: low risk (nodes <5%) → may omit. AUA 2018: unfavourable intermediate or high risk. EAU 2021: nomogram risk **>5%**. AUA/ASTRO 2022: risk-stratified by nomogram.
- **Briganti**/MSKCC nomograms predate mpMRI and PSMA PET.
- **Extended boundaries**: proximal 2 cm above common iliac bifurcation (ureter crossing); distal node of Cloquet; lateral **genitofemoral nerve**; medial bladder wall; all obturator fossa and internal iliac tissue.
- **Clip lymphatics** → fewer lymphoceles. Open and laparoscopic PLND equivalent; node count a poor quality surrogate.

PLND complications (~1 in 7)

- Obturator nerve: clip → remove; transection → primary repair. Ureter at risk proximally — minimise heat.
- **Lymphocele — commonest** (**2–5%** significant). **Transperitoneal access is not protective** (bladder walls off the fossa) → **peritoneal interposition flap** reduces it. Classically ipsilateral DVT.

8 · Postoperative care

- ERAS; same-day discharge in selected men. Drain not routine.

Scenario	Catheter	Cystogram
Routine	~1 week, not before day 5	No
Extensive neck reconstruction / doubtful	≥10 days	Yes
Salvage	2–3 weeks	Strongly consider

- Removal day 3–4 → more retention. Extravasation delays but does not reduce long-term continence.
- Pelvic floor training preoperatively and from ≥3 days after catheter removal.
- **Penile rehabilitation**: PDE5i help on-demand erections, but early rehabilitative use does not improve later response (3 RCTs). Injections for non-responders; vacuum device preserves length.

9 · MIS-specific complications

Complication	Rate	Management
Transfusion	1–2%	Pneumoperitoneum compresses veins
Bladder neck stricture	<2% (open up to 16%)	Dilatation/incision → rarely urethroplasty
Thromboembolism	~0.5%	Pneumatic compression; drugs only with multiple risk factors
Robotic failure	<0.5%	Cancel or convert

Complication	Rate	Management
Recognised rectal injury	Rare; higher in salvage	Multilayer repair + flap, no colostomy
Unrecognised rectal injury	Rare	Rectovesical fistula → colostomy, later repair
Urine leak	Uncommon	Drain, longer catheter, cystogram

- **Small bowel injury** usually off-screen → delayed peritonitis → urgent repair.
- **Ureteric injury**: behind vas (posterior approach); posterior bladder neck after TURP/median lobe; proximal PLND.
- DVT → image pelvis for lymphocele/haematoma; significant lymphocele → duplex for occult DVT.

> **KEY POINT**
> Robotic injuries are fewer but less visible: bleeding and strictures fall with magnified vision, while bowel injury and positioning neuropathy happen outside the camera's field and must be anticipated.

10 · Outcomes

- **Oncologic**: PSM open = robotic (~15%; ~9% organ-confined); driven by tumour (Gleason ≥7, perineural invasion, stage) and surgeon volume — **not** obesity, gland size or prior surgery. BCR = PSA >0.2 ng/mL.
- **Continence**: no-pad mid-90s% at 1 year. Persistent mild → sling or ProACT; severe (<2%) → artificial urinary sphincter.
- **Potency**: up to 90% at 1 year in expert hands; recovery continues to 5 years.
- **Salvage RALRP**: exclude metastases (PSMA PET); full bowel prep; air-bubble test for rectal injury; catheter 2–3 weeks; continence only 33–80%; usually impotent.

11 · Landmark trials

Trial	Design	Result	Take-home
Yaxley 2016	RCT, one open vs one robotic surgeon	Equal continence, potency, early oncology at 3 months; blood loss ~1400 vs 300 mL; shorter stay	Robot wins on recovery, not cancer or function
Retzius-sparing RCTs (Cochrane)	Single-centre RCTs	Earlier continence; equal at 12 months; more PSM	Use selectively

162 Radiation Therapy for Prostate Cancer

- P2 P4 Radiotherapy and brachytherapy for CA prostate; Role of radiotherapy in focal and advanced CA prostate — *α/β, schedules, brachytherapy, ADT by risk, salvage, Phoenix, toxicity*

1 · Radiobiology and fractionation

- Conventional = **1.8–2 gray (Gy)** daily. **Linear-quadratic model**: response to fraction size set by **α/β ratio** — low (2–4 Gy) in late-responding normal tissue, high (≥8 Gy) in most tumours.
- Prostate cancer (few cycling cells): α/β **≈1.5 Gy** (EBRT cohorts 1.0–1.7), **lower than rectum (~3 Gy)** → fractions >2 Gy favour tumour kill over late toxicity = rationale for hypofractionation.

Category	Dose/fraction	Typical course
Conventional	1.8–2 Gy	~8 wk (78 Gy/39)
Moderate hypofractionation	2.25–4 Gy	~4 wk (**60 Gy/20**)
Ultrahypofractionation = SBRT	≥5 Gy	≤5–7 fractions (**36.25 Gy/5**)

2 · EBRT techniques

- mpMRI fused with planning CT (MRI alone lacks electron density). PSMA PET before RT for unfavourable intermediate/high risk changes management in up to half → *Ch 152*.
- **IMRT/VMAT** (inverse planning) vs 3D-CRT: ↓ GI, not GU toxicity (urethra inside target); no RCT, yet standard.

Volume	Prostate application
GTV	Dominant intraprostatic lesion (DIL) on MRI
CTV	Prostate ± proximal seminal vesicles ± pelvic nodes
PTV	CTV + margin: **3–7 mm** with image guidance

- **IGRT** = NCCN standard: daily gold fiducials or cone-beam CT; smaller margins make large fractions safe.
- **Rectal spacer** (hydrogel/hyaluronic acid into Denonvilliers' fascia, transperineal): late rectal toxicity **2% vs 7%** (hydrogel, 79.2 Gy); grade ≥2 2.9% vs 13.8% (hyaluronic acid, 60 Gy/20).
- **MR-linac** (MIRAGE): 2 mm margins ↓ acute GU toxicity.
- **Protons**: Bragg peak shrinks only the low-dose bath, not the high-dose rectal volume; no randomised benefit over IMRT.

3 · Dose escalation

- Six RCTs (MD Anderson, PROG 95-09, Dutch, MRC RT01, GETUG 06, RTOG 0126): ↑ **biochemical control in all risk groups**, **no OS gain** (effective salvage; metastasis only 6% → 4%) → ≥~78 Gy conventional is standard.
- Cost: grade ≥2 late toxicity doubled at 78 Gy; rectal risk when >25% of rectum gets 70 Gy.
- **FLAME**: focal boost to MRI DIL ↑ biochemical PFS (HR 0.45) without more toxicity.

4 · Hypofractionation

- **Moderate**: CHHiP 57/19 not shown noninferior; PROFIT 60/20 noninferior; RTOG 0415 70 Gy/28 noninferior but more late toxicity; HYPRO 64.6 Gy/19 **not superior**. Postoperative HYPORT 62.5 Gy/25 noninferior to 66.6 Gy/37.
- **SBRT**: HYPO-RT-PC (no ADT) ~84% FFS at 5 yr, more acute toxicity.
- **Dose ceiling**: no gain above **40 Gy/5**. Alternate days; margins 3–5 mm; rectal risk if >35% of circumference >39 Gy; spacer ↓ ulcers.

5 · Brachytherapy (BT)

	Low-dose-rate (LDR)	High-dose-rate (HDR)
Source	Permanent seeds (^{125}I, ^{103}Pd, ^{131}Cs)	Temporary ^{192}Ir (380 keV), remote afterloader
Technique	TRUS-guided transperineal template; **peripheral loading** spares urethra	Dwell times/positions; no post-implant dosimetry
Uses	Monotherapy low/intermediate; boost	Boost (e.g. 15 Gy × 1 after 46 Gy); monotherapy 27 Gy/2, 34–38 Gy/4
Pitfall	Oedema, seed migration	**19 Gy × 1 fails**: 5-yr local failure 29% vs 3% (27 Gy/2)

Isotope	Half-life	Monotherapy	Boost (with 45 Gy EBRT)
^{125}I	**59.6 d**	140–160 Gy	108–110 Gy
^{103}Pd	17 d	110–125 Gy	90–100 Gy
^{131}Cs	9.7 d	100–115 Gy	80–85 Gy

- **Post-implant dosimetry** (CT ± MRI, 14–30 days): **D90** >90% of prescription; **V100** >85%; urethral V150 <0.05 cm³; rectal V100 <1 cm³. Low D90 → biochemical failure. MCQ D90 is a dose, V100 a volume.
- **Relative contraindications**: **IPSS >20**; **Qmax <10 mL/s**; large median lobe; large TURP defect or TURP within 3–6 mo; prior pelvic RT; inflammatory bowel/connective tissue disease; **pubic arch interference**; no lithotomy.
- Large gland: ADT shrinks ~30%, 5ARI ~20%. Retention: IPSS >20 ≈ 29% vs <10 ≈ 2%; pre-implant α-blockers.
- **With EBRT**: 45 Gy + implant at 60–70% of monotherapy dose. **RTOG 0232**: BT alone = EBRT + BT in intermediate risk, less toxicity → BT alone standard.
- **ASCENDE-RT**: no OS difference; LDR boost grade 3 GU **18.4% vs 5.2%**.
- UK HDR boost trial: 55 Gy ± HDR 17 Gy/2 → 21% absolute relapse-free gain at 12 yr.

> **KEY POINT**
> Why BT works: the steep fall-off lets intraprostatic dose far exceed EBRT. Hence outcome depends on implant accuracy (D90), eligibility on whether the urethra tolerates swelling, and EBRT is added only where disease may lie beyond the implant.

6 · ADT with RT

- **Synergy**: ADT → apoptosis, smaller target, ↓ **Ku70** (impaired non-homologous end-joining); RT up-regulates AR signalling, which ADT blocks → *Ch 164*.

Risk group (NCCN)	With EBRT	With EBRT + BT	Key evidence
Low / favourable intermediate	None	—	RTOG 94-08: no benefit in low risk
Unfavourable intermediate	**4–6 mo**	—	RTOG 94-08, DFCI 95-096, TROG 96.01 (6 > 3 mo); RTOG 99-10: 9 mo no better than 4
High / very high	**24–36 mo** (≥18)	12–36 mo	EORTC 22863, RTOG 92-02, EORTC 22961 (36 > 6 mo); PCS IV: 18 = 36 mo

- Duration falls as RT dose rises: **TROG 03.04 RADAR** — 18 vs 6 mo benefit lost after HDR boost.
- **STAMPEDE**: + 2 yr abiraterone in high-risk N0/N1 M0 ↑ MFS and OS.

7 · Pelvic nodal RT

- **RTOG 94-13** (node risk >15%): early PFS gain lost; **GETUG-01**: none. Why: L5/S1 border misses **common iliac nodes** (over half of first nodal recurrences) → RTOG 09-24 uses **L4/L5**.
- † **POP-RT** (node risk >20%, long ADT): whole-pelvis to L4/L5 ↑ 5-yr biochemical FFS (~95% vs ~81%) and DFS; no OS difference. More acute GI.

8 · Post-prostatectomy RT

- RADICALS-RT, GETUG-AFU 17, RAVES (64–66 Gy) + meta-analysis → **early salvage preferred**; adjuvant may suit very high risk (Gleason 8–10, pT3b) → *Ch 164*.
- Dose ≥64 Gy; RTOG 96-01 grade ≥2 GU 14%. PSMA, fluciclovine or choline PET localise recurrence.

9 · Follow-up and biochemical failure

- PSA never undetectable. Nadir ~0.6 ng/mL (70 Gy), ~0.3 (79 Gy), ≤0.1 (EBRT + BT); reached ~1.5 yr EBRT, ~2.5 yr BT.

	ASTRO 1996	Phoenix 2005 (current)
Failure	3 consecutive rises	PSA **≥ nadir + 2 ng/mL**
Date	Backdated midway nadir–first rise	When reached (or salvage started)
Notes	Delays ≥18 mo; invalid with ADT/BT	Best correlates with clinical relapse; valid with short ADT; fewer false calls from bounces

- Poor prognosis: failure <18 mo; nadir <12 mo; nadir >2 ng/mL at 1 yr; **PSADT <3 mo** (cancer death; <12 mo → consider early intervention). Local: slow rise, PSADT ≥1 yr; distant: PSADT 3–6 mo.
- **Bounce** (rise >0.2 then durable fall): ~1/3 after LDR (2/3 in young men), up to 12–18 mo, mostly first 3 yr; predicts *better* outcome. After BT: rise within 30 mo → watch; >30 mo → biopsy; approaching 10 ng/mL → systemic staging.
- **Biopsy** optimal at **30–36 mo**. Do not grade marked radiation effect (behaves as negative); high-molecular-weight cytokeratin separates atypia from cancer. Recurrence arises at the original dominant nodule.

10 · Toxicity

Toxicity	Features	Management / prevention
Acute GU	Peak ~2 mo, settle by 12 mo; BT: IPSS curve follows isotope half-life	α-blockers; catheter
Late GU	Grade ≥2 ~20%; stricture; incontinence ~3%, more after TURP	IGRT halves late GU; peripheral loading
Late GI (proctitis)	Bleeding 6–18 mo; IMRT 81 Gy grade ≥2 ~5%; BT 1–2%; colostomy ~0.3%	Steroid/sucralfate enemas → **argon plasma coagulation** or hyperbaric oxygen → colostomy
ED	Vascular, over 24 mo; mostly ADT; penile bulb dose; best with BT alone (~76% at 5 yr)	PDE5i effective in ~¾
Second cancers	† Small excess bladder and rectal cancers	Investigate haematuria or rectal bleeding fully

11 · Salvage after local failure

- Needs **biopsy-proven, local-only** recurrence: salvage RP (incontinence ~63%), BT (preferred), SBRT re-irradiation, cryotherapy, HIFU.

12 · Landmark trials

Trial	Design	Result	Take-home
RTOG 0126	70.2 vs 79.2 Gy, low–intermediate	8-yr biochemical failure-free 65% → 80%; OS same	Dose escalation ↑ control, not OS
CHHiP	74/37 vs 60/20 vs 57/19	60/20 noninferior	60 Gy/20 standard
HYPO-RT-PC	42.7/7 vs 78/39	Noninferior; equal late toxicity	Ultrahypofractionation valid
PACE-B	36.25/5 vs conventional/moderate	Similar 2-yr toxicity	SBRT safe
ASCENDE-RT	EBRT vs LDR boost (+12 mo ADT)	Biochemical failure HR 2.04 for EBRT boost; more GU toxicity with LDR	BT boost for high risk
TROG 96.01	RT ± 3 or 6 mo ADT	6 mo ↑ OS; 3 mo no mortality gain	≥6 mo short-course
RTOG 94-08	66.6 Gy ± 4 mo ADT, T1b–T2b	10-yr OS 57% → 62%; intermediate only	ADT for intermediate, not low
EORTC 22863	RT ± 3 yr goserelin, T3–T4	10-yr OS 39.8% → 58.1%	Long ADT in high risk
EORTC 22961	6 vs 36 mo ADT	Short course not noninferior	Long ADT for high risk
PCS IV	18 vs 36 mo	No survival difference	18 mo acceptable
SPCG-7; NCIC PR.3/MRC PR07	Lifelong ADT ± RT	RT ↓ cancer death, ↑ OS	Never ADT alone if fit
RTOG 94-13; GETUG-01	Pelvic vs prostate RT	No durable benefit	Field height matters
SPPORT	Salvage bed RT ± ADT ± nodes	Pelvic RT + ADT best	Treat nodes in salvage

163 Focal Therapy of Prostate Cancer

1 · Definition and rationale

- **Focal therapy (FT)**: guided ablation of an image-defined, biopsy-confirmed lesion with a **safety margin**, sparing NVBs, sphincter and rectum → sexual and urinary function.
- Model: **breast lumpectomy**. Rests on mpMRI, targeted biopsy and **Gleason pattern 4/5** being the lethal subgroup.

Index lesion hypothesis

- Multifocal in **50–91%** of RP specimens. **Index lesion (IL)** = largest/highest-grade focus; hypothesis: IL drives natural history and metastasis.
- **For**: IL holds **80–90%** of tumour volume; satellites mostly **< 0.5 mL** (Stamey "insignificant") → 52–93% functionally unifocal.
- **Against**: aggressive features in a satellite in 12–38%; IL Gleason matches overall in only ~⅔; a lethal clone traced to a small pattern 3 focus.

> **KEY POINT**
> "Multifocal, therefore unsuitable for FT" is a trap: what matters is one significant lesion plus insignificant satellites, not the count of foci. The hypothesis's exceptions are why FT is always paired with surveillance.

- **FT + AS** = one strategy: ablate grade 4 disease, watch the low-grade remainder, retreat new foci.

2 · Patient selection

- Patient must be **compliant with close surveillance**.
- **TRUS biopsy** misses almost half the disease relevant to hemiablation vs **transperineal mapping biopsy (5-mm grid)**; >half of "TRUS-unilateral" men disqualified.
- **mpMRI** detects high-grade and anterior tumours (15–33%).
- **PROMIS**: csPCa sensitivity **93%**, specificity 41%, NPV 89% → lesions need biopsy; MRI-negative areas still need systematic biopsy.
- mpMRI **underestimates volume** → margin: **5 mm** ablates >95%, 6 mm 98.6% of csPCa. ~**90%** of outfield csPCa lies within **1 cm** of target ("penumbra").
- PSMA PET more specific; mpMRI more sensitive.

> **KEY POINT**
> Imaging finds the lesion, only biopsy grades it, and only systematic biopsy vouches for the rest of the gland — every consensus has kept all three.

Consensus criteria (2016–2019)

Variable	Consensus
Work-up	mpMRI + targeted + systematic biopsy
Risk group	**Intermediate**; Gleason **3+4 = "sweet spot"**; 4+3 maximum; no cribriform or ductal
Size	≤1.5 mL (≤3 mL if not crossing midline); 20–25% of gland if crossing

Variable	Consensus
Residual untreated disease	Gleason 3+3 only (≤1 mm core length in 2016, from 5 mm)
Life expectancy	Per guidelines; no fixed limit
Erectile dysfunction	**Not** a contraindication
Not suitable	High risk (no long-term data); low risk usually → AS

- Goal evolution: eradicate all cancer (2009) → csPCa (2013–16) → treat and retreat to **delay or avoid radical therapy** (2019). Investigational; ideally in a trial.

3 · Ablation patterns and energies

- FT = **lesion, quadrant or hemiablation**; "hockey-stick" (subtotal) is partial-gland ablation, not FT. Transrectal → posterior; transperineal → anterior; transurethral → anterior gland.
- P2 P4 HIFU; Cryotherapy for carcinoma prostate — *HIFU ≥55 °C, posterior lesions; cryo Joule–Thomson, urethral warmer, fistula ≤0.8%*

Modality	Mechanism / route	Pros	Cons
TULSA	Transurethral transducer + MR thermometry	Anterior gland; control within 1.3 mm	In-bore MRI; csPCa on 12-month biopsy ~⅓
Focal laser	Interstitial fibre ≥60 °C; MR thermometry	Most predictable zone	Narrow margin → **highest infield failure** (~40% at 6 yr)
IRE	Non-thermal pulses >1600 V/cm → membrane pores; probes 10–20 mm apart	Spares connective tissue; best function vs matched RARP	Deep anaesthesia (muscle contraction); NVB/urethral injury on histology
VTP	IV padeliporfin (WST11, TOOKAD) 4 mg/kg + 753-nm light → vascular occlusion	**Only randomised evidence** (PCM301)	Gleason 6 only; perineal pain ~15%; light precautions
Radiofrequency	~100 °C	Fat limits conduction	Infield csPCa 20–30%
Focal brachytherapy	LDR seeds or HDR catheters	Mild early toxicity	HDR: stricture ≤8.2%, ED in half; biopsy only at 24 mo

KEY POINT

Every energy fails at the edge of the ablation zone, where temperature or field strength becomes sublethal. Margins, thermometry and adjuvants (cryo-enhancers, SBRT boost) all aim to make that edge lethal. Hence laser's precise but narrow zone gives the most infield recurrence.

4 · Outcomes

Modality	Infield positive biopsy	Outfield	Continence	Erections
Cryotherapy	7–39% overall (~¼ csPCa)		95–100%	47–100%
HIFU	17–54% (csPCa 0–22%)	7–43% (csPCa 2–8%)	90–95%	50–78%
Laser	11–50%	17–60%	95–100%	Minimal change
IRE	csPCa 1–10%	csPCa 5–16%	~All	Some decline
VTP	19–37%	17–19%	Preserved	Nil–mild fall

- Cryotherapy 5-yr BPFS ~62–70%; COLD: partial = whole-gland BPFS, ~2× better erections.
- HEAT (HIFU): 7-yr failure-free survival **69%** (68% intermediate, 65% high risk).

5 · Complications

- Commonest: **urinary retention** (0–9%); UTI, sloughing, stricture (~4–5% HIFU, IRE).
- **Rectourethral fistula** ≤2% — most feared; some heal with catheter 4–6 weeks. ED worst after focal HDR.

6 · Follow-up and failure

- **PSA** falls only **50–80%**; Phoenix/ASTRO not validated after FT. Proposed: nadir + **1.0** ng/mL at 12 months or + 1.5 at 24–36 months predicts csPCa. DRE unreliable.

Success (ICUD-SIU 2015)	Infield	Outfield
Short term	Eradication of treated focus	No csPCa — any outfield cancer = **selection failure**
Intermediate/long term	No csPCa	No csPCa (new cancer may be de novo)

ALGORITHM

mpMRI + targeted biopsy (4–6 cores) of treated zone at **3–6 months** → mpMRI at 1–2 and 5 years (targeted biopsy for new lesion) → systematic biopsy at 1–2 and 5 years for the untreated gland

- **PI-FAB**: read **DCE first** (scar obscures T2/DWI). 1 (DCE−, DWI−) → monitor; 2 (DCE+, DWI−) → biopsy if PSA rising; 3 (both +) → biopsy.

7 · Salvage

- After FT failure: **focal retreatment** or whole-gland RP/RT. † Salvage RP feasible, but harder (fibrosis) and continence/potency worse than primary RP.

Focal salvage for radiorecurrent disease

	Cryotherapy	HIFU	IRE
Cancer control	5-yr BPFS ~50%	3-yr BPFS 48%	PFS 60%, MFS 91% (48 months)
Continence	87–100%	64–87.5%	93%
Signature complication	RUF ~1.4%	Bladder-neck stricture, fistula ~2%; grade ≥3 ~¼	Urethral sludge ~9%
Trial	FORECAST (cryo vs HIFU, ongoing)		SAFE pilot

- Focal ≈ whole-gland salvage for control, better function.

8 · Landmark trials

Trial	Design	Result	Take-home
PCM301 (Azzouzi 2017)	415 low-risk men; padeliporfin VTP vs AS	Progression 28% vs 58%; conversion to radical therapy 4 yr **24% vs 53%**; continence preserved	First level 1 evidence for FT — but Gleason 6 only, surrogate end points

Trial	Design	Result	Take-home
PROMIS	mpMRI vs 5-mm TMB, 11 centres	csPCa sensitivity 93%, specificity 41%	MRI lesions need biopsy; systematic biopsy still needed
HEAT registry	Prospective focal HIFU, 1379 men	7-yr failure-free survival 69%	Largest FT dataset; includes high risk

KEY POINT

Open problems: can insignificant satellites be safely left; cost of repeated treatment; adjuvants. Lumpectomy works because the breast gets adjuvant radiotherapy — prostate FT relies on surveillance and retreatment instead.

164 Treatment of Locally Advanced Prostate Cancer

- P2 P1 Staging and management of locally advanced prostate cancer; Staging and management of non-organ-confined CA prostate — *definition, risk groups, work-up, RT + ADT, RP, salvage RT, pN1, algorithm*

1 · Definition and natural history

- Working definition: **T3–4, N0 or N1, M0 → multimodal therapy**.
- NCCN high risk: cT3a, PSA >20 or GG 4–5. † Very high: cT3b–T4, primary pattern 5, ≥2 high-risk features, or >4 cores GG 4–5.
- Stage migration: high risk now mostly **high grade**, not palpable T3.
- **Grade drives death**: Gleason 8–10 untreated → 15-yr cancer mortality 60–87% (pre-PSA), still >20% today. Swedish cohort: T3–T4 progress like T1–T2 (~48%) but DSS 57% vs 81%.

2 · Staging and risk assessment

- Clinical staging and imaging both **understage**. Partin (localised only); Kattan RP nomogram not for cT3b or N+; CAPRA (each 2-point rise doubles recurrence); Decipher reclassifies ~⅔.
- NCCN staging triggers: cT3–T4, PSA >20, GG 4–5 or primary pattern 5 → bone scan + CT/MRI; PSMA PET may substitute.
- **PSMA PET**: nodal specificity ≥95%, sensitivity only ~40–66%; **proPSMA** accuracy **92% vs 65%**.

3 · Radical prostatectomy in high-risk disease

- **Clinical overstaging of cT3 in 7–27%**; >⅓ of high-risk specimens confined → don't exclude on clinical stage.
- NCCN: RP + **ePLND** only if **completely excisable**; otherwise ADT + RT.
- cT3: ~half recurrence-free at 8–10 yr; DSS **~80–90% at 10 yr**, though biochemical DFS often ~50%.

> **KEY POINT**
> Operative principle: a **positive surgical margin** carries the highest relative risk of progression (4.37) — higher than ECE, SVI or nodes — so remove all prostatic tissue regardless of stage.

Pathology after RP	5-yr PSA-free survival	Relative risk of progression
Organ-confined	95%	Reference
ECE	76%	2.2–2.7
SVI	37%	2.6
Node-positive	18%	3.3
Positive margin	—	**4.37** (highest)

- pT3bN0 at 10 yr: BCR-free ~25%, MFS 70%, DSS 80%. **Node-positive**: single node → nearly half never progress.

Neoadjuvant AD before RP

- PSA ↓ ~90%, volume 30–50%; **clinical downstaging 32–90%, pathological 8–31%**. ↓ margins in cT1–T2, not cT3; **no** effect on SVI, nodes, PSA recurrence, DSS, OS. 8 vs 3 mo: margins 23% → 12%, no PSA benefit.

- **CALGB 90203** (AD + docetaxel before RP): missed primary endpoint, ↑ MFS and OS. pT0 0–4%.

Adjuvant and salvage RT after RP

Trial	Population / comparison	Result
EORTC 22911	pT3 and/or margin+; immediate RT vs observation	bDFS 74% vs 53%; **only with positive margins**
SWOG 8794	Same	**MFS (HR 0.71) and OS (HR 0.72)** — only OS-level evidence
ARO 96-02	pT3N0, undetectable PSA	Biochemical HR 0.53
RADICALS-RT, GETUG-AFU 17, RAVES	Adjuvant vs early salvage (64–66 Gy)	**Early salvage preferred**; 5-yr event-free 87–94% both

- **Trap**: very high-risk (Gleason 8–10, pT3b) were only ~20% of timing trials → adjuvant RT may still help them, as may extensive/multiple margins.
- Dose **≥64 Gy**. **GETUG-AFU 16**: **3 months** AD with salvage RT → 10-yr PFS 64% vs 49% (HR 0.54) → *Ch 162*.

Adjuvant AD and pN1

- **pT3N0**: adjuvant AD ↓ progression, **no DSS/OS gain**. **ECOG 7887** (pN+): immediate AD ↑ OS, DSS, PFS.
- **AD + RT beats AD alone** in pN1: 10-yr DSS 86% vs 70%, OS 74% vs 55%; OS HR ~0.4–0.46. AD alone ↑ other-cause mortality (HR 3.05).
- Adjuvant docetaxel or mitoxantrone after RP: no benefit.

> **ALGORITHM**
> **pN1 after RP**: favourable (few nodes, Gleason ≤8) → observation with PSA monitoring → adverse (>3 nodes, Gleason >8) → AD + RT (best OS) → avoid AD alone where fit.

4 · RT + ADT

- Standard: **dose-escalated RT (>72, ideally ~78–81 Gy) + ADT**, often pelvic RT → *Ch 162*.
- Canadian trial: positive 24-mo biopsy 65% (RT) → 28% (+3 mo NAD) → 5% (neoadjuvant + concurrent + adjuvant).

Trial	Question	Verdict
RTOG 86-10	4 mo neoadjuvant/concurrent AD, bulky T2–T4	Disease-specific mortality 36% → 23%; **OS unchanged**
TROG 96.01	RT ± 3 or 6 mo NAD	6 mo ↓ all-cause mortality (HR 0.63); 3 mo did not
RTOG 85-31	Indefinite adjuvant goserelin	Better control, MFS; OS gain only Gleason 8–10
PCS IV	18 vs 36 mo	OS comparable; 18 mo better QoL
RTOG 94-13	Pelvic vs prostate RT × neoadjuvant vs adjuvant AD	Early PFS gain lost; whole pelvis + adjuvant AD worst OS

- **Duration**: intermediate 4–6 mo; high risk NCCN **18–36 mo** with EBRT, 12–36 mo with brachytherapy boost.

- **ASCENDE-RT**: LDR boost halved biochemical recurrence vs EBRT boost (HR 2.04; 9-yr 85% vs 65%); OS equal. Gleason 9–10: EBRT + boost ↓ cancer mortality (HR ~0.4).
- **Trap**: AD + RT benefits often exclude OS — OS gain with long AD in high risk (EORTC 22863: local control 97% vs 79%; RTOG 92-02 Gleason 8–10 OS 81% vs 71%).

Adding RT to ADT

- **SPCG-7**: cancer death RR **0.44**, overall mortality RR 0.68.
- **PR.3/PR07**: DSS HR 0.54; 7-yr OS 66% → 74%.
- RT adds **8–10% OS**.

Systemic intensification

Trial	Added agent	Verdict
RTOG 0521	Docetaxel to long-term AD + RT	OS HR 0.69
GETUG 12	Docetaxel + estramustine to AD	Relapse-free 62% vs 50%
RTOG 9902	Paclitaxel + estramustine + etoposide	No benefit; thromboembolism
STAMPEDE (high-risk M0/N1)	**Abiraterone + prednisolone 2 yr** to 3 yr AD + RT	**MFS HR 0.53, OS HR 0.60**

- STAMPEDE eligibility: **N1, or two of Gleason ≥8, cT3–4, PSA ≥40**. **NCCN 2023: RT + ADT + 2 yr abiraterone for very high risk.**

5 · ADT alone and other options

- **MRC** immediate vs deferred AD: fewer cancer deaths, mainly M0. Immediate vs deferred orchiectomy (declining local therapy): OS equal.
- **Bicalutamide 150 mg**: better PFS in locally advanced, ↑ **death in localised** (HR 1.23) — never at low risk.
- AD alone does not stop local progression: >half need TURP; ureteric obstruction, haematuria.
- **Cryoablation** vs RT (randomised): equal DSS/OS, worse biochemical control (failure 87% vs 53%).
- **RP vs RT + ADT**: no completed RCT; **SPCG-15** (cT3N0M0) ongoing.
- After RT alone for cT3: 10-yr local recurrence 24–74%, TURP ~⅓, hydronephrosis 20%.

ALGORITHM

Locally advanced / high-risk M0: stage (PSMA PET or CT + bone scan) → fit, resectable → RP + ePLND → adverse pathology: early salvage RT (adjuvant if pT3b/Gleason 8–10/margins+); pN1: observe or AD + RT. Alternative → dose-escalated RT (± brachytherapy boost, ± pelvic nodes) + 18–36 mo ADT → very high risk or N1: + 2 yr abiraterone → unfit for local therapy: ADT alone.

KEY POINT

Viva: "Why not ADT alone for T3 disease?" — SPCG-7 and PR.3 show RT added to ADT halves cancer death and improves OS; AD alone neither cures nor controls the primary, and it carries its own toxicity.

6 · Landmark trials

Trial	Design	Result	Take-home
SPCG-7	ADT ± RT, mostly T3	Cancer death RR 0.44	Add RT to ADT
NCIC PR.3/MRC PR07	Lifelong ADT ± RT	OS HR 0.77	Add RT to ADT
EORTC 22863	RT ± 3 yr goserelin	5-yr OS 62% → 79%	Long ADT in high risk
RTOG 92-02	4 vs 28 mo ADT	OS gain Gleason 8–10	Prolong ADT for high grade
STAMPEDE (abiraterone)	ADT + RT ± 2 yr abiraterone	MFS HR 0.53, OS HR 0.60	Standard for very high risk/N1
SWOG 8794	Adjuvant RT vs observation	MFS, OS improved	Postoperative RT works
RADICALS-RT / RAVES / GETUG-AFU 17	Adjuvant vs early salvage	No difference	Early salvage for most
ECOG 7887	pN+ immediate vs delayed AD	OS improved	Treat pN+ early
proPSMA	PSMA PET vs conventional	Accuracy 92% vs 65%	PSMA PET for staging

165 Management of Recurrent and Newly Metastatic Prostate Cancer

1 · Biochemical recurrence: concept and definitions

BCR: PSA rise after curative therapy, preceding clinical recurrence. Not every detectable PSA progresses, but clinical recurrence is almost always preceded by one.

- After radical prostatectomy (RP): **¼–⅖** develop BCR within 10 years; only ≈half of intermediate/high-risk men with BCR fail clinically.
- MSK: PSA **≥0.4 ng/mL followed by a further rise** best predicted metastasis and secondary therapy.
- **Ultrasensitive PSA** is prognostic, not diagnostic; ≈half of early detectable values fall on retesting → repeat, don't treat.

Setting	Definition	Logic
Post-RP — AUA	PSA **≥0.2 ng/mL**	No gland → any PSA suspect
Post-RP — NCCN	Failure to reach undetectable, two consecutive rises, or PSA ≥0.1	Any of three triggers
Post-RP — EAU	No threshold; risk groups (below)	Kinetics + grade
After EBRT ± ADT	**Phoenix**: **nadir + 2 ng/mL**	Remnant gland makes PSA; validated against metastasis and death
After LDR brachytherapy	≥0.2 may suit better; **<0.2 at 4 years ≈ cure**	≥2× EBRT dose → very low nadir; Phoenix may flatter brachytherapy

EAU BCR risk group	After RP	After RT
Low risk (needs BOTH)	PSADT **>1 year** AND GG <4	Interval to failure **>18 months** AND GG <4
High risk (EITHER)	PSADT ≤1 year OR GG 4–5	Interval ≤18 months OR GG 4–5

KEY POINT

PSA bounce after RT: transient rise of 0.1–0.5 ng/mL at mean ≈ **9 months** (radiation prostatitis, delayed cell death), then a fall. It is **favourable** (better BCR-free survival) → monitor; never salvage on a bounce.

2 · Risk prediction and natural history

- **Positive margin** = only modifiable BCR predictor (pT3a margin-negative beat pT2 margin-positive). Genomic classifiers unproven for selecting salvage.
- **CAPRA** = pre-op; **CAPRA-S** = post-op MCQ; only MSK nomograms reach 15-year cancer-specific survival.
- **Pound**: median **8 years** BCR → metastasis, then **5 years** → death (P2-144).
- After Phoenix failure: ≈5.4 years to metastasis, ≈10.5 to cancer death. Follow-up lifelong (¼ recur after 5 years).

3 · Imaging at recurrence

Purpose: **exclude metastasis before further local therapy**, then localise local recurrence.

- P2 65 yr male underwent RP… PSA has risen — *BCR definitions, PSMA PET/mpMRI work-up, salvage RT dose/timing, RTOG 9601, GETUG-AFU 16, SPPORT, STOMP/ORIOLE, EMBARK, Pound prognosis*
- Bone scan/CT rarely positive below PSA **10 ng/mL**. **PSMA PET**: ≈½ positive at PSA **<0.5**; finds disease outside the salvage field in ≈1 in 5. **EMPIRE-1**: fluciclovine-guided salvage improved 3-year control.
- mpMRI best for local recurrence; after RT, DCE/DWI carry it. **Biopsy after RT** mandatory before local salvage.

> **KEY POINT**
> Salvage RT works best at the lowest PSA, but imaging detection and certainty of recurrence improve only as PSA rises. Risk features (grade, margins, kinetics) let us act early without overtreating.

4 · Salvage after radical prostatectomy

- **SRT** = only curative option (P2-144). Dose **≥64–65 Gy** to fossa; ≈10% RFS lost per 0.1 ng/mL with ≥2 adverse features; IMRT standard.
- No visible lesion on mpMRI predicts SRT failure (occult metastasis).

5 · Salvage after radiotherapy

Prerequisites: biopsy-proven local recurrence, no metastasis on imaging, excellent performance status, life expectancy ≥5–10 years, shared decision. Every modality ≈ **50–70%** control → selection (local-only, low PSA, GG <4) matters more than modality.

Option	Selection / features	Signature morbidity
Salvage RP (open/robotic)	EAU: life expectancy ≥10 years, PSA **<10**, biopsy GG ≤3, N0M0 on imaging, initial cT1–T2; gives pathological staging; poor if SV/nodal involvement	Incontinence, bladder neck contracture, rectal injury, near-universal impotence
Salvage brachytherapy (LDR ≈ HDR)	Only after EBRT	Grade ≥3 urinary toxicity ≈1 in 6
Salvage SBRT	After EBRT	Low toxicity, modest control
Salvage cryotherapy	PSA <10 (ideally <4), PSADT **≥16 months**; post-cryo nadir <0.2 predicts success	**Urethrorectal fistula** (1–3%), ED; incontinence single-digit with **urethral warming**
Salvage HIFU	≥2 years after EBRT	Fistula, osteitis pubis, stricture, incontinence

Recurrence after **focal therapy**: work up like RT failure (PET → MRI → targeted biopsy → any salvage).

6 · Non-curative approach to BCR: observation vs ADT

- AUA: observation (or trial) after maximal local therapy; ADT only for higher metastatic risk — PSADT **<12 months** or high grade; intermittent preferred. EAU: observe low-risk BCR, limited life expectancy or men declining local therapy.
- Primary ADT for localised disease **not** recommended.

- **Messing (ECOG)**: immediate ADT for pN+ after RP improved OS/PFS — but small, pre-modern imaging, bulky nodes, delayed arm treated only at clinical progression. Low-volume pN+ with undetectable PSA: **20–40%** cured by surgery alone.
- MRC: immediate ADT reduced progression-related symptoms; OS gain faded. TOAD (rising PSA) OS HR 0.55; EORTC 30891 benefit only if PSA >50 or PSADT <12 months.
- **EMBARK** entry: PSADT **≤9 months**, PSA ≥1 after RP or ≥2 above nadir after RT.

Intermittent vs continuous ADT

- **NCIC PR.7** (rising PSA after RT): intermittent **non-inferior** (OS 8.8 vs 9.1 years), better QoL.
- **SWOG 9346** (new metastatic): OS 5.1 vs 5.8 years, HR 1.10 (90% CI 0.99–1.23) — **inconclusive** (exceeded 1.20 margin) → intermittent may be worse when metastatic.

7 · Androgen deprivation therapy: biology and agents

Castration removes ligand, not receptor: CRPC keeps AR signalling via AR amplification (20–30%), LBD mutations (antagonist → agonist), splice variants (**AR-V7** — poor prognosis), intratumoural steroidogenesis, GR bypass.

Castrate testosterone: <50 ng/dL.

- P4 P2 GnRH (LHRH) analogues; Maximum androgen blockade and bicalutamide — *agonist flare/FSH rise, antagonists (degarelix, relugolix/HERO), antiandrogen toxicities, CAB, withdrawal response*

Method	Key facts
Bilateral orchiectomy	≈95% of testosterone testicular; castrate within hours (median ≈15 ng/dL). Fewer cardiac events, fractures, PAD than agonists, equal survival → offer for lifelong ADT
LHRH agonists / antagonists	See P4-103. HERO: castrate by day 4 56% vs 0%
Cyproterone acetate	Inferior to goserelin; CV events up to 10% → not for cancer; 50–100 mg/day for hot flushes
ARTAs: enzalutamide 160 mg, apalutamide 240 mg, darolutamide 600 mg twice daily	Seizures → enzalutamide; rash, hypothyroidism → apalutamide; darolutamide fewest effects
Abiraterone (selective CYP17), ketoconazole (non-selective)	Abiraterone ↓cortisol → ↑ACTH → mineralocorticoid excess (↓K^+, HTN, oedema) → give with prednisone. Ketoconazole: castrate within 4 h — emergency option

KEY POINT

Flare matters with spinal/weight-bearing bone metastases: antiandrogen with (or before) the agonist for **21–28 days**; simultaneous start as good as a week's lead-in. **Impending cord compression** or severe pain → **antagonist or orchiectomy** (or ketoconazole). **Antiandrogen withdrawal**: 15–30% respond (bicalutamide); 0–10% with ARTAs; no OS gain.

8 · ADT side effects and their management

Effect	Key facts	Management
Osteoporosis / fracture	Loss of peripheral oestrogen; fracture risk ↑20–45%	Baseline DEXA + FRAX; all: calcium 1000–1200 mg + vitamin D 400–1000 IU/day, exercise, stop smoking. Drug if T-score −1 to −2.5 with 10-year hip fracture **>3%** or major fracture **>20%** (or osteoporosis): denosumab 60 mg SC 6-monthly, zoledronic acid 4–5 mg IV yearly or alendronate 70 mg weekly; repeat DEXA at 1 year
Hot flushes (commonest, 50–80%)	Hypothalamic oestrogen loss	Oestrogens most effective; megestrol, cyproterone, venlafaxine, gabapentin. Clonidine ineffective
Metabolic	Fat ↑, sarcopenia; atypical metabolic syndrome (subcutaneous fat, ↑HDL); diabetes (agonist HR 1.44)	Exercise; monitor glucose, lipids
Cardiovascular	Agonists ↑ CAD, MI, sudden death, VTE	Risk-factor control; orchiectomy/antagonist favourable
Anaemia	Normocytic; most on CAB lose ≥10% Hb within 6 months; pre-ADT Hb <12 g/dL predicts shorter survival	Check blood count; reversible ≈1 year after stopping
Gynaecomastia	≈½ with bicalutamide 150 mg or enzalutamide; ≤5% with castration	Prophylactic breast RT 10–15 Gy, tamoxifen; established → surgery

Follow-up on ADT

3–6-monthly PSA, blood count, chemistry. **Rising PSA → check testosterone first**: inadequate suppression = drug problem (switch or orchiectomy); rising PSA with castrate testosterone = **CRPC** *→ Ch 166* → restage. Germline *HSD3B1* (1245A>C) predicts ADT resistance MCQ.

9 · Metastatic hormone-sensitive prostate cancer (mHSPC)

- P3 Management of newly Metastatic prostate cancer — *CHAARTED/LATITUDE definitions, doublets and triplets with HRs, prostate RT for low volume, MDT, genetic testing, follow-up*
- Prognosis: metachronous low volume best (median OS ≈92 vs ≈43 months synchronous high volume); nodes > bone > visceral. **SWOG 9346 PSA nadir at 7 months**: ≤0.2 → median OS **75** months; 0.2–4 → 44; >4 → 13.

Definition	Criteria	Use
CHAARTED high volume	**≥4 bone metastases with ≥1 outside the axial skeleton**, OR any visceral (no grade criterion)	Docetaxel benefit; RT to primary in low volume
LATITUDE high risk	**≥2 of**: Gleason ≥8, ≥3 bone lesions, visceral	Abiraterone entry

- **ADT is the floor, never the ceiling**; long-term CAB not recommended. Docetaxel **75 mg/m² 3-weekly × 6** (meta-analysis HR 0.77). Observation only if asymptomatic with life expectancy <5 years.

- Prostate RT benefit on top of modern combinations unknown. Cytoreductive prostatectomy: trials or palliation only.
- **MDT**: oligometastases = **≤3 sites** on PET; SBRT or salvage node dissection.

10 · Landmark trials

Trial	Design	Result	Take-home
RADICALS, RAVES, GETUG-AFU 17 (ARTISTIC)	Adjuvant vs early salvage RT (PSA 0.1–0.2)	Equivalent; ≥60% spared RT	Early salvage preferred
RTOG 0534 (SPPORT)	Bed vs bed + 4–6 months ADT vs pelvis + ADT	5-year freedom from progression 70.9 / 81.3 / **87.4%**	Add ADT + pelvic nodes (not OS)
RTOG 9601	SRT ± 2 years bicalutamide 150 mg	12-year OS 76.3 vs 71.3%	OS benefit
GETUG-AFU 16	SRT ± 6 months goserelin	10-year BCR-free 64 vs 49% (HR 0.54); MFS HR 0.73; OS unchanged	Short ADT: no OS gain
EMBARK	High-risk BCR; leuprolide ± enzalutamide vs enzalutamide	MFS 87.3% / 80.0% / 71.4% (leuprolide); combination HR 0.42, enzalutamide alone 0.63	Enzalutamide for high-risk nmHSPC
CHAARTED (ECOG 3805)	ADT ± docetaxel	OS 57.6 vs 44.0 months (HR 0.61); high volume **49.2 vs 32.2** months	High volume only
GETUG-AFU 15	ADT ± docetaxel (up to 9 cycles)	OS HR 0.88, NS; better PFS	Negative docetaxel trial
STAMPEDE (docetaxel; arm G abiraterone)	Multi-arm; metastatic + high-risk M0	Docetaxel OS HR 0.78; abiraterone OS HR 0.63, 3-year OS 83 vs 76%, FFS HR 0.29	Both work
LATITUDE	High-risk mHSPC; ADT ± abiraterone	OS HR 0.62; rPFS 33.0 vs 14.8 months	Abiraterone in high risk
ARCHES	ADT ± enzalutamide	rPFS HR 0.39	Volume-independent
ENZAMET	ADT + enzalutamide vs ADT + older antiandrogen	OS HR 0.67	Control not placebo
TITAN	ADT ± apalutamide	rPFS HR 0.48; OS HR 0.67	Volume-independent
PEACE-1	Synchronous; 2×2 ± abiraterone ± prostate RT	All: rPFS HR 0.54, OS HR 0.82; with docetaxel: rPFS 0.50, OS 0.75	Triplet with abiraterone
ARASENS	ADT + docetaxel ± darolutamide	OS HR 0.68	Darolutamide in mHSPC only with docetaxel
STAMPEDE arm H	ADT ± prostate RT	Overall HR 0.92; low volume HR 0.68 (3-year OS 81 vs 73%)	RT to prostate in low volume

166 Management of Castration-Resistant Prostate Cancer

1 · Definition and biology

- P2 Castration-resistant prostate cancer (CRPC): definition and treatment — *EAU definition, AR mechanisms, evaluation, general principles, nmCRPC, mCRPC options table, sequencing, supportive care*

CRPC = progression (biochemical, radiographic or clinical) despite castrate testosterone **<50 ng/dL**. Historically, ADT alone in metastatic disease: PFS 12–20 months, OS 24–36 months.

- **Castration resistance ≠ androgen independence**: AR still drives (amplification, enhancer amplification, LBD mutations, **AR-V7**, adrenal androgens up to 10%, intratumoural synthesis) → rationale for ARPIs. Once AR dependence is lost → taxanes, radium-223, PSMA, PARPIs, immunity.

2 · Clinical states and progression criteria

TNM does not guide advanced disease; **PCWG3** (2016) clinical states classify by response to prior therapy: localised → rising PSA → nmCRPC → mCRPC first, second, later lines.

- † PCWG3 PSA progression: rise ≥25% and ≥2 ng/mL above nadir, confirmed ≥3 weeks later.
- PSADT **<3 months** → rapid course → consider aggressive therapy.
- Radiographic progression without PSA rise is common with early ARPI use (ARCHES: up to ⅔) → image periodically.

Modality	Pitfall	PCWG3 rule
PSA	Chemotherapy may cause rise before fall; little change with sipuleucel-T/radium-223; flare with radium-223 and ^{177}Lu-PSMA	Confirm rise
Bone scan	Early new lesions may be **flare**	Rescan at **12–16 weeks** as new baseline; progression = **≥2 new lesions** ("2 + 2")
CT	Soft tissue small fraction of burden	RECIST; PFS preferred over response rate
PSMA PET	Apparent progression at 4 weeks resolves by 3 months in ≈13%	Criteria not defined

> **KEY POINT**
> Every marker (PSA, bone scan, PSMA PET) can flare early, so progression is called on confirmed, combined evidence — otherwise effective therapy is abandoned too soon.

- **Continue ADT indefinitely**; exogenous testosterone can cause severe flare.
- Symptomatic bone pain or visceral-predominant disease → **upfront chemotherapy**; limited asymptomatic disease → sequential hormonal therapy.

3 · Germline and somatic testing

- **Germline**: all metastatic; family history of *BRCA1/2-*, Lynch- or *HOXB13*-associated cancers; Ashkenazi ancestry. **Somatic** NGS on tissue (usually archival primary); liquid biopsy alternative but **CHIP** gives false DNA-repair calls (no PARPI response).
- Somatic HRR mutations in **20–25%** of mCRPC; dMMR/MSI-H in **2–5%** → Lynch counselling.

4 · Non-metastatic CRPC (nmCRPC)

Rising PSA on castration, no metastases on **conventional** imaging; shrinking as PSMA PET reclassifies men. PSA kinetics strongest predictor; baseline PSA >10 also predicts; median time to bone metastasis ≈30 months.

ARPI (apalutamide, enzalutamide, darolutamide) + ADT when PSADT ≤10 months — all improve MFS and OS. Abiraterone active (IMAAGEN) but **not approved**.

5 · mCRPC: androgen-signalling inhibitors

Agent	Mechanism	Dose	Toxicity / notes
Abiraterone	Selective CYP17 inhibitor (17α-hydroxylase + 17,20-lyase)	1000 mg **fasting** + prednisone (10 mg/day in mCRPC)	Mineralocorticoid excess, hepatotoxicity; correct K^+ first. Food ↑ absorption up to 10× (250 mg with low-fat meal alternative)
Enzalutamide	Pure AR antagonist; blocks translocation, DNA binding	160 mg, no steroid	**Seizures (≈1%)**, fatigue, falls; CYP3A4/2C8 interactions
Apalutamide	No BBB crossing	240 mg	**Rash, hypothyroidism**, fractures
Darolutamide	Structurally distinct; no BBB crossing	600 mg twice daily	Least CNS toxicity

KEY POINT

Prednisone with abiraterone: CYP17 block ↓cortisol → ↑ACTH → mineralocorticoid excess; prednisone suppresses ACTH. Enzalutamide needs no steroid (prednisone may activate a promiscuous AR). ARPI class: fatigue, falls, fractures — caution >75 years.

6 · Cytotoxic chemotherapy

Only taxanes extend life. **Docetaxel**: β-tubulin → microtubule stabilisation, G2/M arrest; impairs AR nuclear transport. **Cabazitaxel**: poor P-glycoprotein affinity → active after docetaxel.

Agent	Dose / line	Signature toxicity
Docetaxel	**75 mg/m² 3-weekly** + prednisone 5 mg twice daily; first line	Neutropenia, **neuropathy**, oedema, hyperlacrimation, nail dystrophy
Cabazitaxel	**20 mg/m²** (25 if low neutropenic risk) 3-weekly + prednisone + G-CSF; post-docetaxel	**Neutropenia** (grade 3–4 82%), **diarrhoea**
Mitoxantrone	12 mg/m² + prednisone; palliation	No survival benefit
Platinum (+ etoposide or taxane)	Neuroendocrine variant; DNA-repair-deficient	—

Estramustine abandoned; adding anti-angiogenic/bone drugs to docetaxel never improved OS.

7 · Radiopharmaceuticals

Feature	Radium-223	^{177}Lu-PSMA-617
Emitter	**Alpha** (path **<100 µm**, spares marrow); calcium mimetic	Beta; needs PSMA PET-positive disease

Feature	Radium-223	^{177}Lu-PSMA-617
Eligible	**Symptomatic bone metastases, no visceral or bulky nodal disease**	PSMA-avid, any site, after ARPI + taxane
Schedule	50 kBq/kg IV 4-weekly × 6	6-weekly, up to 6 cycles
Toxicity	Diarrhoea, anaemia	Fatigue, **dry mouth**, anaemia
Caution	**Never with abiraterone/prednisone**; bone agent with any ARPI	—

Strontium-89, samarium-153 only palliate pain.

8 · PARP inhibitors

- P4 PARP inhibitors in Prostate cancer — *synthetic lethality, PARP trapping, AR–PARP synergy, BRCA2 vs ATM, approvals, toxicity, key trials*
- Ineffective in unselected men. ATM-deficient tumours may suit ATR inhibitors.

9 · Immunotherapy

- **Sipuleucel-T**: autologous dendritic-cell vaccine (PAP–GM-CSF fusion protein); first approved therapeutic cancer vaccine. For **asymptomatic/minimally symptomatic mCRPC, no visceral metastases or opioid pain**. Prolongs OS **without** PSA/PFS change. Toxicity: infusion fever, rigors.
- Checkpoint inhibitors fail in unselected mCRPC ("cold" tumour). **Pembrolizumab** 200 mg IV 3-weekly: tumour-agnostic for **MSI-H/dMMR** or TMB **≥10 mutations/megabase** after ≥1 line.

10 · Neuroendocrine/aggressive-variant CRPC

Feature	Usual mCRPC	Neuroendocrine variant
AR / PSA	Retained; PSA tracks disease	Lost; **PSA low despite progression**
Pattern	Osteoblastic bone ± nodes; visceral <20%	Bulky masses, liver, **osteolytic** lesions with hypercalcaemia (PTHrP), brain
Molecular	AR alterations	≥2 of *PTEN* loss, *TP53* mutation, *RB1* loss; low PSMA
Treatment	ARPI, taxane, radioligand, PARPI	**Platinum + etoposide** (or taxane + carboplatin) ± RT
Survival	Years	**<12 months**

Biopsy a metastasis if rapid progression with low PSA; chromogranin A not needed for diagnosis.

11 · Sequencing and cross-resistance

Sequential ARPI helps only **15–30%** → **switch mechanism** (CARD). Sequencing table in P2-140; add pembrolizumab for MSI-H/dMMR, PARPI if HRR-mutated.

- Enzalutamide induces CYP3A4/2C9/2C19 (anticoagulants, antiepileptics).
- † AR-V7 in circulating tumour cells predicts resistance to abiraterone/enzalutamide, not taxanes.

12 · Complications and palliative care

- P2 Metastatic prostate cancer with bone metastases — *cord compression pathway, Patchell, pain RT, obstruction, zoledronic acid vs denosumab, ONJ prevention*

Every focal back pain in a man with bone metastases is cord compression until proven otherwise; may arise from paravertebral soft tissue (normal bone scan).

> **ALGORITHM**
> Back pain ± deficit or sphincter change → **high-dose IV dexamethasone** (10 mg IV, then 4–10 mg 6–12-hourly; 16–100 mg/day) → **urgent whole-spine MRI** → RT ± decompressive surgery → taper steroids over 2–3 weeks

- Operate if progression during RT, unstable fracture, or recurrence after RT.
- Neuropathic pain → tricyclics, gabapentinoids, nerve block. Circumoral numbness → skull-base metastasis → irradiate.
- Obstructive uropathy → **relieve before chemotherapy**; not all unilateral cases need drainage.
- Anaemia: erythropoiesis-stimulating agents not recommended. Start hospice care in time.

13 · Bone-targeted therapy

(1) ADT osteoporosis; (2) **SRE prevention** in CRPC bone metastases — **no survival effect**.

Feature	Zoledronic acid	Denosumab
Dose	**4 mg IV 4-weekly**, renal adjustment	**120 mg subcutaneous (SC) 4-weekly**; no renal adjustment
SRE	Better than placebo	**Better than zoledronic acid** (20.7 vs 17.1 months)
Toxicity	Flu-like, creatinine rise, hypocalcaemia, ONJ	Hypocalcaemia (grade ≥3 5%), ONJ **2–4%**

> **KEY POINT**
> **ONJ**: mostly with dental extraction/poor dentition → dental assessment first. Calcium 1000 mg + vitamin D 800 IU daily with both. Bone agents strongly recommended with radium-223.

14 · Landmark trials

Trial	Design	Result	Take-home
SPARTAN	nmCRPC, PSADT ≤10 mo; apalutamide	MFS 40.5 vs 16.2 mo (HR 0.28); OS HR 0.78	First nmCRPC drug
PROSPER	nmCRPC; enzalutamide	MFS improved; death ↓27%	Enzalutamide for nmCRPC
ARAMIS	nmCRPC; darolutamide	MFS 40.4 vs 18.4 mo; OS HR 0.69	Least toxic
TAX 327	Docetaxel 3-weekly vs weekly vs mitoxantrone	OS 18.9 vs 17.3 vs 16.4 mo (HR 0.76)	First chemo to prolong survival
TROPIC	Post-docetaxel; cabazitaxel vs mitoxantrone	OS 15.1 vs 12.7 mo (HR 0.70)	Second-line chemo

Trial	Design	Result	Take-home
PROSELICA / FIRSTANA	Cabazitaxel 20 vs 25 / vs docetaxel first line	20 non-inferior / not superior	20 mg/m^2; docetaxel first
CARD	After docetaxel + 1 ARPI: cabazitaxel vs 2nd ARPI	OS 13.6 vs 11.0 mo (HR 0.64)	Switch mechanism
COU-AA-301	Post-docetaxel; abiraterone	OS 14.8 vs 10.9 mo (HR 0.65)	Post-chemo
COU-AA-302	Chemo-naive, asymptomatic	rPFS 16.5 vs 8.3 mo	Pre-chemo
AFFIRM	Post-docetaxel; enzalutamide	OS 18.4 vs 13.6 mo (HR 0.63)	Post-chemo
PREVAIL	Chemo-naive; enzalutamide	Death ↓29%; radiographic progression ↓81%	Pre-chemo
ALSYMPCA	Symptomatic bone-only; radium-223	OS 14.9 vs 11.3 mo (HR 0.70); SRE delayed	Alpha emitter prolongs life
VISION	Post-ARPI + taxane, PSMA+; ^{177}Lu-PSMA	OS 15.3 vs 11.3 mo (HR 0.62); rPFS HR 0.40	Approved 2022
TheraP	Phase II; ^{177}Lu-PSMA vs cabazitaxel	PSA response 65 vs 37%; fewer grade 3–4; OS equal	Lutetium ≥ cabazitaxel
IMPACT	Sipuleucel-T vs placebo	OS 25.8 vs 21.7 mo (HR 0.78); no PSA/PFS change	Vaccine approved 2010
PROfound	Post-ARPI; olaparib vs 2nd ARPI	*BRCA/ATM* rPFS 7.4 vs 3.6 mo (HR 0.34)	Olaparib, HRR panel
TRITON2/3	Rucaparib after ARPI	*BRCA* rPFS HR 0.50; no *ATM* benefit	*BRCA* only
PROpel / MAGNITUDE / TALAPRO-2	First-line PARPI + abiraterone / + enzalutamide	rPFS gains, largest in *BRCA*; PROpel OS 42.1 vs 34.7 mo (P = 0.054)	HRR/*BRCA*
Denosumab vs zoledronic acid	Bone-metastatic CRPC	First SRE 20.7 vs 17.1 mo; OS equal	Denosumab for SRE

★ Putting it together

A · PSA to diagnosis

ALGORITHM

1. Raised PSA or abnormal DRE → repeat; exclude infection, instrumentation, 5-ARIs (×2 PSA); density, biomarkers (*→ Ch 154*).

2. mpMRI before biopsy → PI-RADS (*→ Ch 155*).

3. Biopsy: PI-RADS 3–5 targets + systematic; transperineal safer.

4. ISUP Grade Group (*→ Ch 156*); risk group (*→ Ch 157*); PSMA PET (or CT + bone scan) from unfavourable intermediate.

B · Localised: risk → treatment

NCCN: favourable = ALL criteria, adverse = ANY one; weigh life expectancy, comorbidity, preference.

Risk group	Defined by	Management
Very low	T1c, GG1, PSA <10, density <0.15, <3 positive cores (≤50% each)	AS
Low	T1–T2a, GG1, PSA <10	AS preferred
Favourable intermediate	One factor (T2b–c, GG2, PSA 10–20); <50% cores	RP or RT; AS (low-volume GG2); focal in trials
Unfavourable intermediate	2–3 factors, or GG3, or ≥50% cores	RP, or RT + 4–6 months ADT
High / very high	T3a, GG4–5 or PSA >20 / T3b–T4, primary pattern 5, ≥2 high-risk features	RP + ePLND, or RT + ADT 18–36 months; + abiraterone 2 years if N1 or ≥2 of Gleason ≥8, cT3–4, PSA ≥40 (STAMPEDE)

KEY POINT

SPCG-4, PIVOT, ProtecT: radical treatment helps only younger men, ≥intermediate risk, long life expectancy. ProtecT: RP = RT for 15-year cancer survival.

C · RP vs RT vs focal

	RP (Ch 160–161)	RT (Ch 162)	Focal (Ch 163)
Main toxicity	Incontinence, ED	Rectal/urinary irritation; late ED; ADT	Least; retreatment common
Advantages	Pathological staging; salvage RT	No anaesthesia; comorbid; hypofractionation	Preserves function; MRI-visible index lesion
Failure	PSA ≥0.2 ng/mL (confirmed)	Phoenix: nadir + 2 ng/mL	Undefined; biopsy-based
Salvage	Bed RT, early at PSA 0.1–0.2 ± ADT	RP, cryo, HIFU, brachytherapy	Repeat focal, whole-gland, RP, RT

D · Advanced disease

ALGORITHM

1. BCR (Ch 165): PSMA PET. Post-RP: salvage RT. Post-RT: biopsy-proven → local salvage. High risk (PSADT ≤9 months) → ADT + enzalutamide (EMBARK).

2. mHSPC: never ADT alone: + ARPI; triplet with docetaxel if fit, high volume. Prostate RT if low volume (STAMPEDE arm H). Genetic testing all.

3. nmCRPC PSADT ≤10 months: ARPI (SPARTAN, PROSPER, ARAMIS).

4. mCRPC (Ch 166): switch mechanism: docetaxel → cabazitaxel (CARD); ^{177}Lu-PSMA (VISION); radium-223 symptomatic bone-only (ALSYMPCA); PARPI if BRCA/HRR mutation (PROfound); pembrolizumab if MSI-H. Bone protection always.